

Master's Thesis

by

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**Explain Reality: Grounded Procedural Tutoring
with Foundation Models
and Visual Fact Extraction in Immersive Simulation**

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Abstract

Many safety- and skill-critical domains, including aviation, industrial maintenance, medicine, and emergency operations, require learners to practise ordered procedures whose next correct action depends on the current system state. As such training moves into high-fidelity and immersive simulation, tutoring systems must provide contextual help without relying on language-model fluency alone. This thesis studies that problem through the F/A-18C cold-start procedure in DCS World and designs an evidence-grounded tutoring runtime that integrates procedure-pack-driven state tracking, retrieval-augmented help generation, and a fine-tuned vision-language model (VLM) for structured visual fact extraction from cockpit displays. A central finding is that visual fact ontology design governs model reliability: evolving from an 8-fact to a 13-fact ontology by decomposing compound targets into locally verifiable visual atoms reduced critical false positives by 64–89% on independent holdout sets. A Qwen3.5-9B LoRA adapter trained on 928 bilingual rows achieves 0.99 fact accuracy on two holdout sets with verified zero training overlap; the identical data recipe transfers to Gemma-4-31B with consistent gains, suggesting the approach is not tied to a single backbone. A small formative VR pilot study with nine participants provides descriptive evidence: all four tutor-condition participants completed the full 33-step procedure, whereas none of the five without-tutor participants did; omission errors were largely replaced by assistance-coupled errors in the tutor condition. Controlled evaluation with larger samples remains future work.

Zusammenfassung

Prozedurales Lernen in hochrealistischer Simulation erfordert das Ausführen geordneter, zustandsabhängiger Handlungsabfolgen bei gleichzeitiger Überprüfung von Systemzuständen. Herkömmliche Trainingshilfen unterbrechen jedoch entweder die Immersion oder haben keinen Zugriff auf den Simulatorzustand. Diese Arbeit begegnet dieser Lücke durch eine telemetriegestützte Tutoring-Laufzeitumgebung, die paketbasierte Zustandsverfolgung, Retrieval-Augmented-LLM-Hilfegenerierung und ein feinabgestimmtes Vision-Language-Model (VLM) zur strukturierten Extraktion visueller Fakten aus Cockpit-Anzeigen kombiniert. Ein zentrales Ergebnis ist, dass das Design der visuellen Fakten-Ontologie die Modellzuverlässigkeit bestimmt: Die Weiterentwicklung von einer 8-Fact- zu einer 13-Fact-Ontologie durch Zerlegung zusammengesetzter Ziele in lokal überprüfbare visuelle Atome reduzierte kritische False-Positives um 64–89% auf unabhängigen Holdout-Sets. Ein Qwen3.5-9B LoRA-Adapter, trainiert mit 928 bilingualen Datenzeilen, erreicht eine Fact-Genauigkeit von 0,99 auf zwei Holdout-Sets mit verifizierter Null-Überlappung zum Training; dieselbe Datenrezeptur überträgt sich mit konsistenten Verbesserungen auf Gemma-4-31B, was darauf hindeutet, dass der Ansatz nicht an ein einzelnes Backbone gebunden ist. Eine kleine formative VR-Pilotstudie mit neun Teilnehmern liefert deskriptive Evidenz: Alle vier Teilnehmer mit Tutor absolvierten die vollständige 33-Schritt-Prozedur, während keiner der fünf Teilnehmer ohne Tutor dies schaffte; Auslassungsfehler wurden in der Tutor-Bedingung weitgehend durch assis-tenzgekoppelte Fehler ersetzt. Eine kontrollierte Evaluation mit größeren Stichproben bleibt zukünftige Arbeit.

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Declaration on the Use of AI Tools

AI-assisted tools, including ChatGPT/Codex and Claude Code, were used during the preparation of this thesis for language editing, restructuring suggestions, LaTeX consistency checks, citation-gap auditing, and inspection of repository artefacts under the author's direction. These tools were not used to fabricate experimental data, participant results, or bibliographic metadata. All claims, numerical results, citations, interpretations, and final wording remain the responsibility of the author.

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1 Introduction

1.1 Motivation and Problem Context

Many safety- and skill-critical domains require people to learn ordered procedures whose next correct action depends on the current state of a system. Examples include aircraft cockpit procedures, industrial maintenance, emergency operations, and medical or technical equipment workflows. In such settings, procedural knowledge is not only a list of actions. It also includes preconditions, intermediate checks, state-dependent branching, and recovery from partially completed or incorrectly completed steps.

High-fidelity and immersive simulation make these procedures available for practice without exposing learners to the risks and costs of the real environment. At the same time, fidelity creates a tutoring problem. A learner who is already inside a simulated cockpit, machine, or procedure room may need help that is specific to the current state rather than a general explanation of the task. If the learner has to leave the simulation to consult a manual, video, or checklist, the training flow is interrupted. If the learner instead asks a language model for help, the answer may be fluent but disconnected from the state that actually exists in the simulation.

This thesis studies that problem through the F/A-18C cold-start procedure in DCS World. The flight-simulation setting is used as a concrete and demanding case study, not as a claim that the research problem exists only in aviation. The case is useful because it makes the evidence problem visible: procedure progress depends on switch states, engine parameters, warning lights, cockpit display pages, and the order in which earlier actions were completed. Some of these states are directly observable through telemetry; others require visual inspection of cockpit displays; and a small number require manual confirmation or lie outside the display layout used by the vision pipeline.

The current procedure pack contains 33 ordered steps (S01–S33) grouped into six operational phases, from initial electrical power through engine start, avionics configuration, flight-control checks, and final pre-taxi configuration (Section 2.1). The benchmark and VLM evaluation reported in this thesis focus on the original 25-step core procedure;

the later extension to S33 adds further pre-taxi checks without changing the central evidence problem. This makes the cold-start a compact but representative case for state-dependent procedural tutoring in high-fidelity simulation.

1.2 Research Gap and Problem Statement

Existing training aids and recent foundation-model systems each address part of the problem but leave an important gap. Manuals, videos, and checklists provide authoritative procedural knowledge, yet they do not observe the learner’s current simulator state. Conventional simulator telemetry can expose many states, yet it does not by itself explain what the learner should do next or interpret display content that is not exported as data. Foundation models can generate flexible explanations, but their output is not reliable enough for procedural guidance unless it is constrained by explicit evidence.

The central premise of this thesis is that procedural assistance becomes more reliable and more auditable when every actionable tutoring output is tied to an explicit evidence source. In the system developed here, such evidence may be a telemetry variable, a procedure-gate evaluation, a retrieved manual or checklist snippet, or a VLM-derived visual fact. The language model is therefore not asked to infer procedure state from parametric knowledge alone. It operates inside a runtime that supplies state evidence, checks structured outputs, validates overlay targets, and records the resulting help cycle for replay and analysis.

This framing changes the role of the foundation model. The model is not the tutor by itself; it is one component inside an evidence-grounded procedural tutoring runtime. The research question is therefore architectural and methodological as much as it is generative: how should simulator telemetry, procedure rules, retrieval, and visual fact extraction be connected so that generated help remains state-aware, inspectable, and bounded by available evidence?

1.3 Research Questions

The thesis is guided by one main research question and three subordinate questions:

Main RQ. How can an evidence-grounded multimodal tutoring system support state-dependent procedural training in high-fidelity immersive simulation?

RQ1 — System architecture and evidence integration. How can simulator telemetry, retrieved procedural knowledge, and visual fact observations be integrated so that tutoring outputs are state-aware, auditable, and safe for procedural guidance?

RQ2 — VLM adaptation and ontology design. To what extent can a small-data LoRA-adapted VLM reliably extract structured cockpit visual facts, and how does visual fact ontology design affect critical false positives?

RQ3 — Formative VR pilot evidence. In a VR-based F/A-18C cold-start training task, what formative evidence does a small pilot study provide about task completion, procedural errors, and tutor interaction in a with-tutor versus without-tutor comparison?

RQ1 and RQ2 are addressed through the implemented runtime, replay and harness infrastructure, and technical benchmark evaluation. RQ3 is addressed through a completed formative VR pilot study ($n = 9$). The pilot is treated as descriptive evidence about feasibility, instrumentation, completion, and error patterns; it is not used to claim controlled learning gains, workload reduction, retention, or transfer.

1.4 Approach and Scope

The thesis addresses the research questions by building and evaluating an evidence-grounded tutoring runtime for the F/A-18C cold-start case. The approach combines four evidence channels. First, DCS-BIOS telemetry provides the most authoritative signal for many cockpit controls, lights, and numerical engine or system values. Raw UDP frames are normalised, sanitised, and aggregated into stable runtime variables so that procedure gates can distinguish between a false value and a missing source. Second, where telemetry is insufficient, a visual fact layer converts cockpit display captures into structured facts with states `seen`, `not_seen`, or `uncertain`. Third, retrieved procedural sources provide textual support for explanations. Fourth, procedure-pack gates define which evidence is required before a step can be considered available or complete.

The help cycle is deliberately constrained. The runtime assembles an evidence packet containing the current procedure candidates, telemetry state, visual observations, and retrieved sources before model generation occurs. The LLM/VLM output is then validated against runtime-generated JSON Schema constraints and mapped through allowlisted overlay targets. If the model path is unavailable or invalid, the system falls back to deterministic, text-only guidance. This design makes the help cycle auditable: a later

replay can inspect which evidence was available, which step was inferred, what the model returned, and how the response was validated before display.

The scope is intentionally bounded. The thesis validates the architecture and visual fact pipeline technically, and it reports a completed formative VR pilot study. It does not claim that the system has established controlled learning gains, workload reduction, retention, transfer, or generality across all procedural domains. The F/A-18C case is used to make the evidence-grounding problem concrete and measurable. Extending the approach to other aircraft, industrial equipment, medical training tasks, or emergency procedures would require new procedure packs, evidence mappings, and domain-specific validation.

Within that boundary, the thesis scope is threefold:

1. The design and implementation of an evidence-grounded tutoring runtime that combines DCS-BIOS telemetry, procedure-pack gates, retrieved procedural knowledge, VLM-derived visual facts, structured response validation, overlay display, logging, and replay (Chapters 3, 4, and 5);
2. The technical validation of the visual fact adaptation pipeline through independent holdout benchmarks and cross-backbone comparisons (Chapter 6, Part A);
3. A completed formative VR pilot study ($n = 9$) that provides descriptive evidence about task completion, error patterns, tutor interaction, and instrumentation readiness under with-tutor and without-tutor conditions (Chapter 6, Part B).

Procedure definitions, gates, telemetry mappings, visual fact bindings, and error taxonomies are kept in declarative procedure-pack configuration rather than embedded in prompts. This pack-based design keeps the architecture aligned with Clean / Hexagonal principles: domain rules remain separate from simulator adapters, vision adapters, model adapters, and export tooling.

1.5 Contributions

This thesis makes three contributions.

1. **An evidence-grounded procedural tutoring architecture.** The thesis presents a ports-and-adapters tutoring runtime in which each actionable help output must be traceable to telemetry, a procedure-gate evaluation, a retrieved source, or a

visual fact. The contribution is not merely that an LLM is connected to a simulator. It is that generated tutoring is placed inside a constrained help cycle with declarative procedure packs, schema-validated responses, allowlisted overlay targets, deterministic fallbacks, and replayable logs. This contribution addresses RQ1 and is described in Chapters 3 and 5, with validation evidence in Chapter 6, Section 6.2.

2. **A visual fact ontology and small-data VLM adaptation pipeline.** The thesis develops a 13-fact visual ontology for F/A-18C cockpit display state, evolved from an 8-fact baseline that exposed the weakness of asking a VLM to recognise compound procedural states directly. The revised ontology decomposes such targets into locally observable visual atoms and leaves procedural interpretation to downstream rules. The associated data pipeline spans DCS screenshot capture, AI pre-labelling, human review, bilingual SFT row generation, LoRA fine-tuning, and independent holdout benchmarking. On the Qwen backbone, a LoRA adapter trained on 928 bilingual rows reaches approximately 0.99 fact accuracy on two independent holdouts and reduces critical false positives by 64–89%; the same data recipe also improves the Gemma backbone. This contribution addresses RQ2 and is described in Chapters 3 and 4, with evaluation in Chapter 6, Sections 6.1.2 and 6.1.3.
3. **A VR procedural training evaluation framework and formative pilot study.** The thesis delivers the instrumentation required to study evidence-grounded procedural tutoring with real participants: experiment export tooling, baseline and tutor-condition configurations, session launch support, replayable logs, semi-automated pre-scoring, error coding guides, and an experiment operation manual. A formative VR pilot study ($n = 9$) was conducted with four with-tutor and five without-tutor participants. All four tutor-condition participants completed the full 33-step procedure; none of the five without-tutor participants did. The observed error patterns also differed: tutor-condition errors were mainly assistance-coupled, while without-tutor errors were mainly omissions. These results support feasibility and instrumentation claims, but not controlled claims about learning effectiveness. This contribution addresses RQ3 and is documented in Chapter 6, Section 6.3, and Appendix 9.

Taken together, these contributions define the boundary of the thesis. The work demonstrates that evidence-grounded tutoring can be engineered, benchmarked, and

exercised in a formative VR study for a demanding simulator procedure. It does not claim that educational effectiveness has been established beyond that formative evidence.

1.6 Thesis Structure

Chapter 2 explains why the F/A-18C cold-start task requires multiple evidence sources, introduces the visual fact ontology, and positions the thesis against related work in intelligent tutoring, simulator instrumentation, state-aware assistance, VLM adaptation, and grounded AR/VR systems.

Chapter 3 presents the evidence-grounded architecture, including the ports-and-adapters structure, runtime data flow, procedure and gating model, telemetry processing, replay infrastructure, and VLM component.

Chapter 4 describes the methods used to create and validate the evidence chain: telemetry grounding, knowledge retrieval and source policy, visual fact data collection, ontology evolution, LoRA fine-tuning, and holdout benchmark design.

Chapter 5 documents the implementation choices that make the architecture operational, including core domain logic, adapter boundaries, structured help generation, overlay mapping, VLM integration, and experiment tooling.

Chapter 6 evaluates the thesis claims in two parts: technical evidence for RQ1 and RQ2, followed by the formative VR pilot evidence for RQ3.

Chapter 7 interprets the results, draws design lessons from the ontology revision, discusses scope boundaries, and states the limitations and future work.

Chapter 8 summarises the completed work against the research questions and re-states the claim boundaries of the thesis.

Appendix 9 contains supplementary figures, benchmark tables, dataset statistics, hyperparameter configurations, CLI references, and the 8-fact baseline benchmark.

2 Background and Related Work

The previous chapter framed the thesis around one requirement: generated procedural help should be grounded in evidence that can be inspected before and after it reaches the learner. This chapter explains why that requirement arises in the F/A-18C cold-start task and how it relates to existing work. The first part introduces the task, the available evidence channels, and the visual fact ontology used to turn cockpit screenshots into structured observations. The second part positions the thesis against intelligent tutoring systems, simulator-based training, rule-based state tracking, VLM adaptation, and grounded assistance in AR/VR.

2.1 The F/A-18C Cold-Start Task

The procedural scenario studied in this thesis is the F/A-18C cold-start in DCS World, a pre-flight cockpit procedure that moves the aircraft from a cold and dark state to a taxi-ready configuration [7, 8]. The task is useful as a research case because it is both bounded and state-sensitive. It has a known order of actions, yet successful completion depends on intermediate states that cannot be reduced to a text checklist: electrical power must be established before engine start, engine parameters must enter nominal ranges before the second engine is engaged, cockpit displays must show the expected pages, and some checks only make sense after earlier system transitions have occurred.

For a tutoring system, the relevant property of the cold-start is therefore not only its length. It is the way the procedure forces the tutor to combine three kinds of evidence: sequence evidence about which steps have been completed, telemetry evidence about cockpit controls and instruments, and visual evidence about display-page content. A response that ignores any of these signals can be locally fluent but procedurally wrong.

2.1.1 Procedure Structure: S01–S33 and Phases P1–P6

The current procedure pack represents the cold-start as 33 numbered steps (S01–S33) grouped into six phases. The phase structure is used in the thesis as a compact descrip-

tion of the evidence demands imposed by the task:

- P1 (S01–S03):** Power-up and safety checks. Battery and generator activation, fire-detection tests, and APU start establish the first electrical and safety preconditions for the rest of the procedure.
- P2 (S04–S07):** Right engine start. Right engine crank, throttle advance, bleed-air cycling, and caution/warning/advisory light testing introduce telemetry-dependent timing and verification.
- P3 (S08–S09):** Displays and communications. The DDIs, AMPCD, HUD, display pages, and COMM1 preset configuration make display state part of the procedural evidence chain.
- P4 (S10–S14):** Left engine and core systems. Left engine start, INS alignment mode selection, radar power, and OBOGS activation depend on earlier engine and avionics state.
- P5 (S15–S27):** FCS and flight controls. FCS reset and BIT, flap/trim configuration, launch-bar and arrestor-hook checks, refuelling probe checks, and pitot heat configuration mix telemetry-visible controls with display-based confirmation.
- P6 (S28–S33):** Pre-taxi instruments and final checks. Parking brake, BINGO fuel, altimeter settings, standby attitude indicator uncaging, and attitude-source selection complete the taxi-ready state.

Each step is represented in the procedure pack with precondition and completion gates. These gates make explicit which previous steps and which state variables must be satisfied before the tutor can treat a step as available or complete. For example, the right engine crank step depends on APU support, while the left engine start depends on right-engine parameters such as RPM, temperature, fuel flow, nozzle position, and oil pressure entering nominal ranges. This is why the task is a useful test case for evidence-grounded help: a tutor that advances the learner based on checklist order alone can miss the actual aircraft state.

2.1.2 Cockpit Displays and the Composite Panel

Many cold-start states are visible on the F/A-18C's cockpit displays rather than directly available as simple telemetry variables. The visual pipeline therefore operates on a fixed composite panel image containing three display regions:

- **Left DDI (Digital Display Indicator):** used during the procedure for pages such as FCS, SUPT, and TAC. In the evaluated cold-start flow, the FCS page and FCS page markings are important visual confirmations.
- **AMPCD (Advanced Multipurpose Color Display):** used for the HSI page, map layer, and INS alignment text. These states are central to the alignment-related visual facts.
- **Right DDI:** used for the BIT root page and FCS-MC BIT sequence, including intermediate, in-test, and final GO states.

Using a composite image keeps training and inference aligned: the model sees the same three-display layout during dataset construction, benchmarking, and runtime capture. More importantly, the composite panel defines a narrow perception boundary. The VLM is not asked to interpret the whole VR scene or infer the pilot’s intention. It only reports whether specific cockpit-display facts are visible in known regions.

2.1.3 DCS-BIOS Telemetry

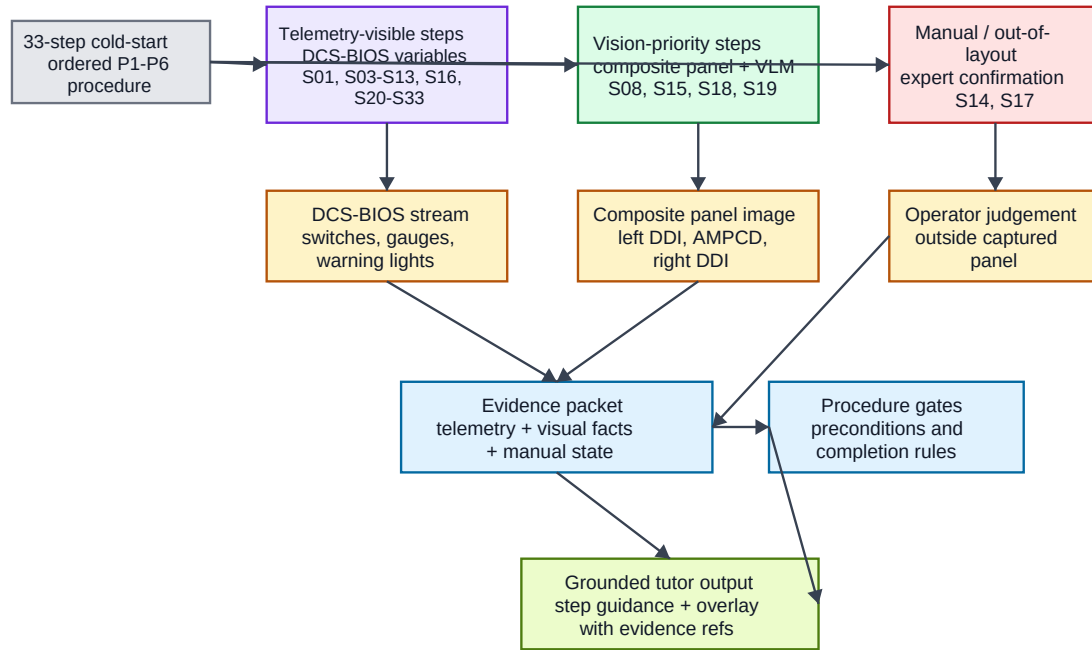
DCS-BIOS provides the main non-visual evidence source for the runtime. It exports cockpit switch positions, indicators, warning lights, gauges, and other instrument states from DCS World through a UDP telemetry stream [6]. The tutoring runtime normalises this stream into stable variables such as `rpm_r`, `apu_ready`, and `ins_mode`, applies delta sanitisation to reduce jitter, and tracks whether a source is missing rather than merely false.

This telemetry channel is authoritative where it exists. Steps such as battery power, throttle movement, radar mode, and many final pre-taxi switch states can be checked without computer vision. At the same time, DCS-BIOS does not export all display-page content needed by the cold-start tutor. The runtime therefore uses vision only for states that telemetry cannot observe, such as whether the FCS page, BIT root page, FCS-MC BIT result, or INS alignment text is visible. Figure 2.1 summarises this division of evidence across the 33-step procedure.

2.2 Visual Fact Ontology

The visual fact ontology is the contract between perception and procedural reasoning. Rather than asking a VLM to produce a tutor decision such as "the learner should press

F/A-18C cold-start evidence sources



The task cannot be grounded by language alone because completion evidence comes from mixed sources.

Figure 2.1: Evidence sources for the F/A-18C cold-start task. The 33-step procedure combines telemetry-visible steps, vision-priority display checks, and a small number of manual or out-of-layout confirmations; downstream tutoring decisions are grounded in the corresponding evidence source rather than in language-model output alone.

this button next", the runtime asks it to produce a small set of structured propositions about the visible cockpit display state. Each fact is a discrete observation with one of three values:

seen: the specified visual evidence is clearly present;

not_seen: the specified visual evidence is absent or not discernible;

uncertain: the image quality, occlusion, crop, or display state is insufficient for a confident decision.

This representation intentionally avoids mixing perception with task interpretation. The VLM reports whether visual evidence is visible. The procedure engine later combines those facts with telemetry, sequence state, and step bindings to decide whether a step is confirmed, pending, or blocked. The ontology is therefore not a minor implementation detail; it defines what kind of claim the model is allowed to make.

2.2.1 From 8-Fact v1 to 13-Fact v2

The ontology evolved through two generations. The first version contained eight facts over the left DDI, right DDI, and AMPCD. It included useful display-level targets such as FCS page visibility, BIT page visibility, FCS-MC page state, and INS alignment page visibility. It also included compound targets such as `ins_go`, which asked the model to infer a high-level procedural completion state from a single visual frame.

That design produced a clear lesson. The first LoRA adapter, trained on 180 human-reviewed Run-001 images, improved in-distribution behaviour but produced critical false positives on the independent Run-002 holdout: 13 of 50 samples were predicted as `seen` for `ins_go` when the human label was `not_seen`. The error was not only a model failure. It exposed a design problem in the target itself: `ins_go` combined visual perception with procedural interpretation.

The second-generation ontology replaces such compound targets with 13 observable visual atoms:

Region	Visual fact
Left DDI	tac_page_visible
Left DDI	supt_page_visible
Left DDI	fcs_page_visible
Left DDI	fcs_page_x_marks_visible
Right DDI	bit_root_page_visible
Right DDI	fcsmc_page_visible
Right DDI	fcsmc_intermediate_result_visible
Right DDI	fcsmc_in_test_visible
Right DDI	fcsmc_final_go_result_visible
AMPCD	hsi_page_visible
AMPCD	hsi_map_layer_visible
AMPCD	ins_grnd_alignment_text_visible
AMPCD	ins_ok_text_visible

The important change is not just the larger number of labels. The ontology moves from procedure-level judgement to observable evidence. For example, the v1 `ins_go` target is decomposed into AMPCD text and map-layer observations shown in the table above. The procedure engine then decides how these observations relate to the alignment step. This design principle carries the main scientific insight of the VLM part of the thesis: VLMs should extract local visual evidence, while procedural meaning should be assigned by explicit downstream rules.

2.2.2 Sticky Facts and Step Bindings

Two runtime mechanisms make the ontology usable during a live help cycle.

Sticky facts. Some visual events are transient but procedurally important. For example, the `fcsmc_final_go_result_visible` fact is marked as sticky with a time-to-live of 600 seconds. Once it has been observed as **seen**, later frames marked **not_seen** or **uncertain** do not immediately erase the evidence. This prevents a completed visual event from disappearing merely because the learner changed pages, the display flickered, or the capture window missed the relevant moment. Non-sticky facts, such as page visibility facts, expire quickly and are re-evaluated continuously.

Step bindings. Visual facts become procedurally meaningful only through explicit bindings in `vision_facts.yaml`. A binding specifies which facts must be visible for a step to be visually confirmed. Some bindings require all listed facts, such as S08 requiring both the FCS page on the left DDI and the BIT root page on the right DDI.

Others allow one of several alternatives when multiple visual cues can support the same procedural state.

The procedure pack also classifies each step by its preferred evidence source:

- `bios_priority_steps` for telemetry-visible states;
- `vision_priority_steps` for display-dependent states;
- `manual_or_out_of_layout_steps` for states requiring manual confirmation or lying outside the three-display composite.

This classification is the practical expression of the thesis’s evidence-grounding principle: each step is checked against the most reliable available signal before it is used in tutoring.

2.3 Intelligent Tutoring Systems and Procedural Training

Intelligent Tutoring Systems (ITS) provide the closest educational precedent for this work. Classic ITS research distinguishes expert knowledge, learner state, and tutoring feedback, and has shown that step-based computer tutors can be effective in structured learning domains [21]. In procedural settings, constraint-based and example-tracing tutors model expected actions and compare learner behaviour against authored task knowledge [2, 17]. These ideas directly inform the use of procedure packs and gates in this thesis.

The difference is the observation interface. Many ITSs operate inside a tutor-controlled environment where learner actions are already represented as formal events. The simulator used here is not a tutor-native interface. The runtime must infer procedure state from DCS-BIOS telemetry, cockpit screenshots, sequence history, and occasional manual confirmations. This makes evidence integration part of the tutoring problem itself.

LLMs add a further tension. They can produce fluent and flexible instructional dialogue, and recent educational discussions recognise their promise as well as their reliability risks [13]. For cockpit procedures, fluency is not enough. The contribution of this thesis is therefore not an open-ended LLM tutor, but a constrained tutoring runtime in which generated language is downstream of evidence collection, retrieval, procedure inference, and validation.

2.4 Simulator-Based Training and Instrumentation

Simulator-based aviation training is a mature area, and prior work has examined how simulation supports skill acquisition and transfer in flight-training contexts [11]. DCS World provides the high-fidelity simulation environment used in this thesis [7]. The relevant research issue for the present work is instrumentation: how the tutoring system observes what the learner and aircraft are doing without replacing the simulator with a custom tutor interface.

DCS-BIOS addresses part of this problem by exposing cockpit state through a structured telemetry bridge [6]. Compared with screen capture alone, telemetry gives deterministic access to many switch positions, indicator states, and numerical values. However, telemetry does not cover every piece of evidence that matters for the cold-start, especially display-page content and text. The thesis therefore treats simulator instrumentation as a multimodal problem: telemetry is used where it is reliable and direct, while vision is reserved for the cockpit evidence that telemetry does not export.

This separation also improves auditability. A logged help cycle can later show which state was read from DCS-BIOS, which facts came from the VLM, which rule made a step eligible or blocked, and which retrieved source supported the generated explanation. Instrumentation is thus not only an engineering requirement but a condition for evaluating the tutor’s behaviour.

2.5 Rule-Based and State-Aware Tutoring

The tutoring runtime combines deterministic state tracking with model-generated language. Rule-based approaches to procedure tracking, including finite-state, production-rule, and constraint-oriented models, offer interpretability because they make task structure explicit. The procedure pack in this thesis follows that principle: steps, preconditions, completion gates, visual bindings, and error categories are authored as declarative configuration rather than hidden in prompts.

The limitation of a purely rule-based tutor is that authored feedback can become brittle when the learner asks for help in an unexpected state or needs an explanation rather than a binary correction. The runtime therefore delegates natural-language formulation to an LLM, but not the authority to decide the procedure state. Step inference, eligibility, evidence availability, and overlay allowlists remain under deterministic control.

This hybrid division is related to broader work on grounding language models in action feasibility and symbolic structure, such as SayCan-style grounding of language in robotic affordances [1]. The present thesis adapts that grounding intuition to simulator tutoring: before a generated response can be acted upon, it must be tied to evidence that the runtime can observe, validate, and replay.

2.6 Foundation Models and VLMs for Situated Assistance

Foundation models provide two capabilities that are attractive for situated tutoring: language models can produce flexible explanations, and vision-language models can extract information from cockpit images. Parameter-efficient fine-tuning methods, especially Low-Rank Adaptation (LoRA), make it practical to adapt large models to narrow tasks with limited trainable parameters [12, 16]. The model backbones and VLM literature used in this thesis provide the general technical context for this adaptation setting [4, 5, 15].

The thesis uses these capabilities in a restricted way. The VLM is not asked for free-form cockpit descriptions, and the LLM is not asked to invent the next procedural state. Instead, the VLM outputs structured JSON visual facts, and the help-generation model receives an evidence packet assembled by the runtime. This choice follows the same general motivation as retrieval-augmented generation: model output should be conditioned on explicit external evidence rather than parametric memory alone [14].

The cockpit-display task also resembles structured screen understanding. Research on GUI and screen-content understanding has shown that visual models can reason about fixed regions, labels, and symbolic layout [23]. Cockpit displays share that structured quality, but the tutoring setting adds stricter consequences for false positives. A VLM that incorrectly reports a critical display state as visible may cause the tutor to advance the learner at the wrong time. This is why ontology design and critical-false-positive analysis are treated as central methodological issues in this thesis rather than as secondary benchmark details.

2.7 Explainable and Grounded AI Assistance in AR/VR

The original proposal placed the tutor in an immersive Quest 3 cockpit experience, and the implemented system retains that evaluation direction through the VR pilot. Prior work on AR guidance for maintenance and procedural tasks shows the value of presenting instructions in the user’s task context, often through spatially registered text, arrows, or animations [18]. Such systems demonstrate why in-context guidance is attractive: the learner can keep attention on the task environment rather than switching to external materials.

The system studied here differs from pre-authored AR guidance because its help content is generated at runtime from current evidence. That makes explanation and traceability central. Work on explainable AI and human-AI interaction emphasises that users and developers need to understand why a system produced a recommendation, especially in high-stakes or safety-sensitive settings [3, 9]. In this thesis, the explanation need is operationalised through evidence references: each recommended step, overlay target, or refusal to advance can be traced back to telemetry, a rule, a visual fact, or a retrieved source.

This grounding requirement also limits the claims made about VR. The pilot study in Chapter 6 shows that the runtime and instrumentation can be exercised with real participants in VR, but it does not establish controlled educational effectiveness. The VR contribution is therefore part of the evaluation framework and formative evidence, while the primary thesis claim remains the design and validation of the evidence-grounded tutoring runtime.

2.8 Positioning of This Work

The thesis sits at the intersection of the areas reviewed above, but its contribution is the particular integration rather than membership in any single area. From intelligent tutoring systems it takes the need for explicit task models and state-sensitive feedback. From simulator training it takes the need for instrumentation that observes real learner behaviour in a high-fidelity environment. From rule-based assistance it takes deterministic procedure gates. From foundation-model work it takes flexible language generation and domain-adapted visual extraction. From explainable and AR/VR assistance it takes the requirement that help remain inspectable in context.

The resulting system is an evidence-grounded procedural tutoring runtime. Its pro-

cedure pack defines what counts as valid state evidence; DCS-BIOS supplies telemetry where available; the visual fact ontology supplies structured cockpit display observations where telemetry is missing; retrieval supplies procedural sources; and the help cycle constrains model output through validation, allowlists, fallbacks, logging, and replay. This is the research position developed in the rest of the thesis.

The claim boundary is equally important. The technical chapters and benchmarks evaluate whether the architecture and visual fact pipeline can produce reliable, auditable evidence for tutoring. The formative VR pilot provides descriptive evidence about feasibility, completion, error patterns, and instrumentation with real participants. Larger controlled studies are still required before making claims about learning gains, workload reduction, retention, or transfer.

3 System Architecture

This chapter answers RQ1 at the architectural level: how can simulator telemetry, retrieved procedural knowledge, and visual observations be integrated so that tutoring outputs remain state-aware, auditable, and safe for procedural guidance? The central design decision is to treat foundation-model output as one stage in an evidence-constrained runtime rather than as the tutor itself. Every help response is produced from an explicit evidence packet, checked against procedure-pack rules, validated by a harness, mapped through an overlay allowlist, and recorded for replay.

The chapter first derives the design requirements from the evidence-grounding problem. It then presents the ports-and-adapters architecture, the live help cycle, the telemetry and procedure-gate evidence layers, the harness validation mechanism, the simulator adapter boundary, logging and replay infrastructure, and the runtime role of VLM-derived visual facts. Dataset construction and VLM fine-tuning methodology are deferred to Chapter 4; implementation-level details appear in Chapter 5.

3.1 Design Requirements

Evidence-grounded procedural tutoring imposes requirements that are different from a conventional checklist application or an unconstrained LLM chatbot. The runtime must observe state, reason over procedure order, explain help in language, and still prevent unsupported overlay actions from reaching the learner. Six requirements guide the architecture.

- **G1 – Live procedural state observation.** The tutor must observe learner progress through simulator state rather than through self-report alone. In the F/A-18C case, this means using DCS-BIOS telemetry where available and visual facts where cockpit display state is not exported as structured data.
- **G2 – Procedure-aware step inference.** The tutor must reason over preconditions, completion gates, and sequence history. The next useful help item is not

necessarily the next checklist line; it is the next action that is valid given the current evidence.

- **G3 – Explicit evidence for actionable output.** Every recommended step, overlay target, or refusal to advance must be traceable to evidence: telemetry, a gate result, a retrieved procedural source, a recent state change, or a visual fact. Model fluency alone is not acceptable evidence.
- **G4 – Clean architectural boundaries.** Domain logic must remain independent of simulator APIs, model providers, GPU availability, and VR display plumbing. This permits the core procedure and validation logic to be tested without a running simulator.
- **G5 – Graceful degradation.** When model output is unavailable, malformed, or insufficiently grounded, the runtime must fall back to deterministic text guidance or safe rejection rather than executing an unsafe overlay.
- **G6 – Declarative procedure configuration.** Procedure steps, gates, telemetry mappings, UI targets, visual fact bindings, and error taxonomies must be represented as a procedure pack rather than hidden inside prompts or adapter code. This keeps the task model inspectable and versionable.

These requirements explain why the system is designed as an evidence pipeline. The architecture does not try to make the language model internally reliable enough to act alone. Instead, it surrounds generation with explicit state evidence, declarative rules, schema validation, allowlists, and replayable logs.

3.2 Ports-and-Adapters Architecture

The runtime follows a ports-and-adapters architecture with a strict dependency direction: external systems connect through adapters, adapters implement stable ports, and the core contains the procedure and validation logic. The core never imports DCS-specific, provider-specific, or VR-specific modules. This boundary is central to the thesis claim because evidence-grounding must be testable and inspectable without depending on the live simulator.

Core layer. The core contains simulator-independent domain logic: procedure state tracking, gate evaluation, visual fact state merging, BM25 retrieval primitives,

event types, response schema generation, evidence packet construction, harness validation, scoring, and replay-oriented state checks.

Ports layer. Ports define the stable interfaces between the domain core and the outside world. The main ports cover model inference, knowledge retrieval, visual input, and telemetry input. These ports make it possible to replace a model provider, knowledge backend, or simulator adapter without changing the procedure engine.

Adapters layer. Adapters contain external integration code: DCS-BIOS receivers, telemetry sanitisation and aggregation, DCS overlay dispatch, OpenAI-compatible and local model clients, BM25 document retrieval, screenshot capture, and VLM visual fact extraction.

Procedure packs. The procedure pack is the task-specific contract consumed by the core. The F/A-18C pack defines step order, precondition gates, completion gates, UI target identifiers, telemetry variable mappings, visual fact bindings, and error taxonomy. It is not an adapter because it does not talk to the simulator; it is the declarative domain specification.

Figure 3.1 summarises this architecture. The diagram should be read from left to right as an evidence flow: simulator state and cockpit images enter through adapters, procedure-pack rules structure the runtime interpretation, retrieved sources and model services support help generation, and the harness checks the resulting action plan before any overlay or tutor text reaches the cockpit.

The same boundary also supports the research workflow. Technical claims about gating, evidence packets, harness decisions, and replay can be tested with fixtures. Claims about visual fact extraction can be benchmarked separately from live tutoring. Claims about VR presentation can be treated as formative pilot evidence rather than as a prerequisite for validating the architecture.

3.3 The Evidence-Grounded Help Cycle

The live help cycle is the core runtime calling chain. In compact form, it is:

help trigger → telemetry snapshot → optional vision capture → step and gate inference →
evidence packet → retrieval → model response → harness validation → overlay/text output
→ event log

Evidence-grounded tutoring architecture

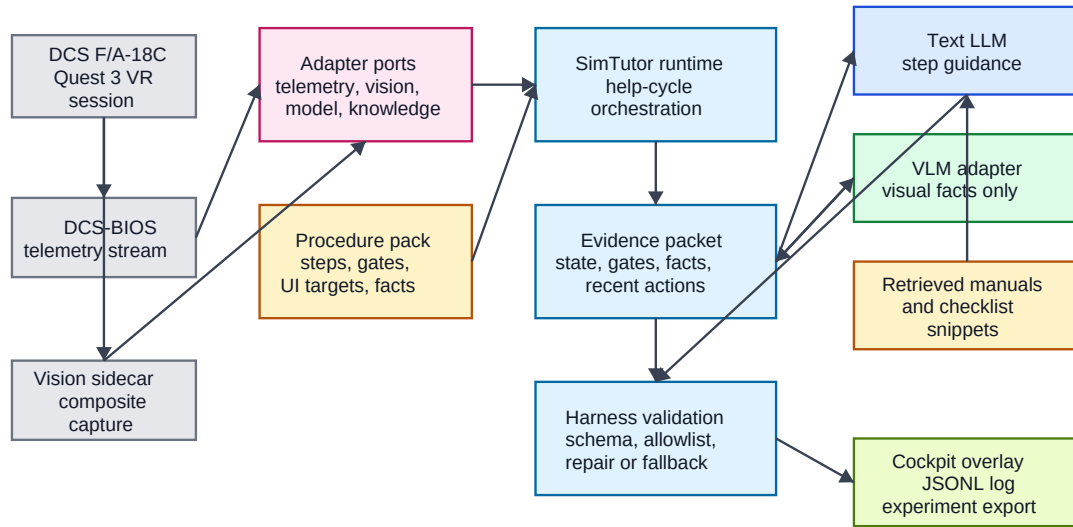
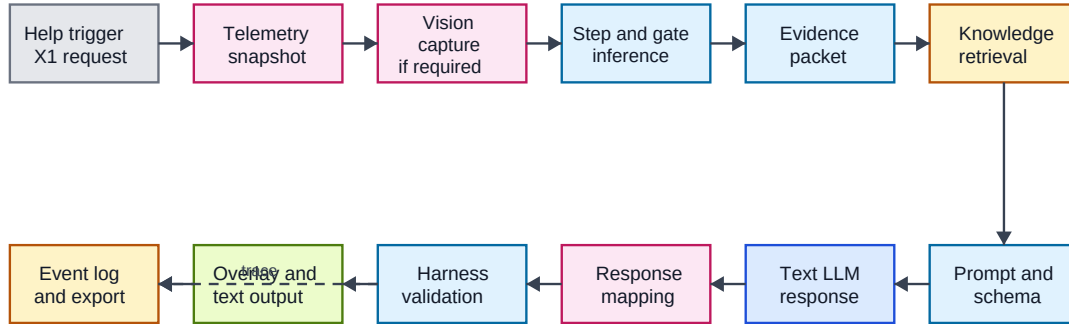


Figure 3.1: Evidence-grounded tutoring architecture. Simulator telemetry, composite-panel vision, procedure-pack configuration, retrieved knowledge, and model outputs are integrated through an evidence packet and checked by the harness before any overlay or tutor text reaches the cockpit.

This sequence matters because it fixes the order of authority. The model does not first decide what the simulator state means. Instead, the runtime assembles state evidence and candidate steps before generation, and validates the generated response before display. Figure 3.2 shows this cycle.

Live evidence-grounded help cycle



The LLM never acts directly on the simulator; mapped responses pass through harness validation first.

Figure 3.2: Live evidence-grounded help cycle. Each help request is converted into a telemetry and vision observation, interpreted against procedure gates, grounded with retrieved knowledge, passed through the text LLM, validated by the harness, and then dispatched as overlay/text output with an event-log trace.

3.3.1 Help Cycle Stages

Observation. The DCS-BIOS adapter receives simulator state and the telemetry pipeline normalises it into runtime variables. When the active or likely step requires display evidence, the vision adapter can capture the composite cockpit panel and request visual fact extraction.

Candidate inference. The procedure engine evaluates precondition and completion gates, identifies candidate steps, and records missing or uncertain evidence. The result-

ing candidates are not natural-language suggestions; they are structured runtime objects derived from the current evidence.

Evidence packet assembly. The runtime collects telemetry values, gate results, recent deltas, visual fact states, retrieved source references, and step candidates into an evidence packet. This packet is the boundary object between state interpretation and model-assisted help generation.

Model response. The prompt asks the model to produce a structured help response grounded in the packet. The output must conform to the runtime `HelpResponse` schema. Free-form text that cannot be parsed or mapped safely does not become an action.

Validation and mapping. The response passes through JSON extraction, schema validation, evidence reference checks, step and target allowlists, and harness validation. Only then can a response be mapped to overlay actions or tutor text. Failed validation triggers repair, rejection, or deterministic fallback.

Execution and logging. Overlay commands are limited to safe display actions such as highlight, pulse, or clear. Control injection and auto-clicking are outside the architecture. Every help cycle is logged as structured events so that the decision can be replayed after the session.

3.3.2 Schema-Level Evidence Grounding

Evidence grounding is enforced before output execution. The `HelpResponse` schema is generated from the current procedure pack and UI map, so the set of valid steps and overlay targets is task-specific. Each target proposed by the model must be accompanied by evidence whose type and reference correspond to data available in the help-cycle context. The supported evidence kinds are telemetry variables (`var`), gate evaluations (`gate`), retrieved sources (`rag`), recent state changes (`delta`), and visual facts (`visual`).

This design changes the failure mode of an unsupported model response. If the model proposes a plausible but ungrounded overlay, the response is not merely logged as questionable; it is structurally invalid for action execution. The system can then repair to an allowed target, fall back to text-only guidance, or reject the action plan.

3.4 Telemetry as State Evidence

Telemetry is the primary evidence source for controls, indicators, and numerical state values that DCS-BIOS exports directly. The telemetry pipeline turns raw simulator

frames into stable state evidence through four stages:

1. **Raw ingestion.** DCS-BIOS UDP frames are decoded into structured observations with timestamps and sequence information.
2. **Variable resolution.** Procedure-pack mappings project raw BIOS fields onto normalised variables such as `rpm_r`, `apu_ready`, or `battery_on`. The resolver also tracks missing sources so the runtime can distinguish an observed false value from an unavailable value.
3. **Delta sanitisation.** Configured thresholds, debounce windows, and filtering policies suppress spurious changes from sensor jitter, switch bounce, and out-of-order frames.
4. **Delta aggregation.** Recent state changes are compressed into a prompt-safe summary and an evidence source for help-cycle reasoning.

The missing-source distinction is important for procedural tutoring. A gate should not fail in the same way when a switch is known to be off and when the telemetry source is absent. The first case can justify corrective guidance; the second may require uncertainty-aware or text-only help.

3.5 Procedure Engine and Evidence Gates

The procedure engine maintains state over the ordered cold-start sequence. Each step can be pending, active, done, or blocked, and transitions are driven by precondition and completion gates. Gates are expressed declaratively in the procedure pack rather than embedded in prompts. This ensures that procedure state remains a rule-governed domain decision rather than a model inference.

The gate evaluator supports threshold checks, range checks, boolean flags, and elapsed-time conditions. Missing, source-missing, or sentinel values produce diagnostic failures instead of being silently coerced into valid state. These diagnostics feed both the evidence packet and deterministic fallback.

3.5.1 Deterministic Fallback

The deterministic fallback is part of the architecture, not an emergency afterthought. When no model is available or when model output fails validation, the runtime computes

a conservative hint from the active step, unsatisfied gates, recent actions, and evidence availability. It distinguishes hard blockers, where evidence shows that a gate is unsatisfied, from soft blockers, where the required evidence is missing or uncertain. Overlay output is suppressed unless a local rule can uniquely determine a valid target.

3.5.2 Step Classification by Evidence Source

The current F/A-18C procedure pack classifies steps according to the most reliable available evidence source:

- **Vision-priority steps:** S08, S15, S18, and S19 require cockpit display evidence from the visual fact pipeline.
- **BIOS-priority steps:** S01, S03–S13, S16, and S20–S33 are declared as primarily telemetry-verifiable in the pack metadata.
- **Manual or out-of-layout steps:** S14 and S17 require manual confirmation or involve cockpit areas outside the three-display composite.

The fire-test step S02 is handled through its own gate and recent-action contract rather than through the pack’s three priority lists. The important architectural point is that the runtime does not ask the model to decide which evidence channel is authoritative for a step. That choice is declared by the procedure pack and consumed by the help cycle.

3.6 Harness Engineering and Evidence Validation

The harness is the architecture’s final safety boundary before output reaches the simulator. It converts a model-proposed response into a validated action plan, or into a repaired, rejected, or text-only outcome. The detailed object relationships are shown in Appendix Figure 9.1; the main architectural role is summarised here.

The harness uses four conceptual objects:

EvidencePacket. A normalised snapshot of the current help cycle: telemetry state, gate diagnostics, recent actions, visual facts, retrieved references, and candidate step information.

StepCandidate. A candidate explanation of where the learner may be in the procedure, derived from gates, telemetry, recent actions, and visual evidence.

StepHarnessSpec. A pack-derived contract for each step: allowed overlay targets, required observables, telemetry facts, vision facts, completion predicates, recovery policy, and signal freshness requirements.

HarnessActionPlan. The validated result: selected step, selected overlay step, targets, evidence references, guidance text, text-only status, repair status, rejection status, and reasons.

The validation sequence has three effects. First, it prevents wrong-step or wrong-target outputs from bypassing the procedure pack. Second, it prevents stale or inappropriate evidence from driving a later step. Third, it creates machine-readable reasons for repairs and rejections, which are then available for replay and evaluation.

3.6.1 Fixture-Based Regression Testing

The harness is tested with live help fixtures: recorded telemetry snapshots and runtime contexts captured at procedure states that previously exposed failures. Each fixture is replayed through the help cycle and asserts the expected outcome, such as the correct target, a repaired target, text-only fallback, or safe rejection.

The current coverage matrix contains 20 live-fixture regression entries across S01–S33. These fixtures cover correct advancement, wrong-target rejection, repair of incomplete or mismatched visual evidence, prevention of stale VLM facts leaking into non-visual steps, moving-control wait guidance, and interaction semantics for switches, push buttons, and rotary controls. This coverage is not an educational evaluation; it is technical evidence that evidence-grounding constraints can be enforced systematically at the runtime boundary.

3.7 Adapter Boundary: Simulator, Overlay, and Vision

The adapter layer isolates DCS-specific behaviour behind the ports used by the core. Three adapter responsibilities are central to the live runtime.

Telemetry input. The telemetry adapter receives DCS-BIOS state data and feeds the telemetry pipeline. Its output is normalised state evidence rather than simulator-specific frame data.

Overlay output. The overlay adapter maps abstract tutor targets to DCS overlay identifiers and sends highlight, pulse, or clear commands. It does not execute cockpit controls.

Vision input. The vision adapter triggers screenshot capture and passes composite-panel images to the VLM fact extractor. The resulting visual facts return to the core as structured evidence.

The VR path follows the same boundary. VR mirror display, tutor text transport, and composite-panel capture are adapter concerns; procedure inference and harness validation remain in the core. The VR code path was exercised in the formative pilot study reported in Section 6.3. Larger controlled human-subject evaluation remains future work.

3.8 Event Logging and Replay

Auditability requires that help-cycle decisions survive beyond the live session. The runtime records versioned, timestamped events in JSONL format. Events include observations, help requests, evidence summaries, model outputs, harness decisions, overlay actions, and scoring-relevant interaction data.

Replay uses these records to reconstruct step ordering, state-machine consistency, and help-cycle behaviour offline. The current replay evaluation suite includes five scenario types: no-operation, battery-on, battery-and-generator-off, FCS reset and BIT, and INS alignment. These cases use DCS-BIOS raw captures and, where available, vision sidecar artifacts.

The logging and replay layer supports the thesis in two ways. During development, it makes regressions reproducible. During evaluation, it connects claims about readiness and instrumentation to concrete artifacts rather than to manual observation alone. Full end-to-end replay of multimodal human-in-the-loop sessions with synchronised vision frames remains a limitation.

3.9 VLM Adapter for Visual Evidence

The VLM component supplies visual evidence for display states that telemetry does not export directly. Its runtime output is a set of visual facts, not a procedure decision. Each fact is a bounded proposition about visible cockpit display content, with state

`seen`, `not_seen`, or `uncertain`. The procedure engine and harness decide how those facts affect the help cycle.

The current runtime contract uses the 13-fact ontology introduced in Chapter 2. Facts have time-to-live settings, optional sticky semantics, and step bindings through all-of or any-of conditions. For example, facts related to FCS-MC final GO can be latched so that a transient completed display state is not lost when the learner changes pages.

When a help request concerns a vision-priority step, the runtime may trigger a capture, extract visual facts, merge them into the current evidence snapshot, and include them in the evidence packet. If the vision adapter is unavailable or the multimodal request fails, the help cycle continues with telemetry, retrieval, and procedure gates. This graceful degradation preserves the telemetry-grounded core while allowing visual evidence to improve steps that need display interpretation.

The data pipeline, ontology evolution from 8 facts to 13 facts, LoRA fine-tuning configuration, and holdout benchmark protocol are methodological questions rather than architecture questions. They are therefore treated in Chapter 4 and evaluated in Chapter 6.

3.10 Summary

The architecture presented in this chapter turns foundation-model tutoring into a constrained evidence-processing pipeline. Telemetry and visual facts provide observations, procedure packs define valid state transitions and evidence requirements, retrieval supplies source-grounded context, models generate structured help, the harness validates or repairs proposed actions, and logs make the decision replayable. This is the system-level answer to RQ1: procedural help becomes auditable when model output is made downstream of explicit evidence and upstream of enforced validation.

4 Methodology

Chapter 3 presented the runtime architecture that keeps tutor output downstream of evidence and upstream of validation. This chapter describes how that evidence is produced, curated, and evaluated. The method is organised around a practical question: how can simulator telemetry, retrieved procedure knowledge, and VLM-derived visual facts become auditable inputs for a procedural tutor?

The chapter has three roles. First, it defines the runtime evidence interfaces used by the procedure engine: telemetry variables, gating diagnostics, retrieved snippets, and deterministic fallback. Second, it presents the visual fact methodology for RQ2: screenshot capture, pre-labelling, human review, ontology revision, bilingual SFT export, LoRA adaptation, and holdout benchmarking. Third, it documents the formative VR pilot protocol for RQ3. The pilot is treated as descriptive evidence about feasibility and interaction patterns; it is not used for inferential claims about learning gains.

4.1 Procedure Runtime and Telemetry Grounding

Telemetry and procedure gates provide the method-level state evidence on which the help cycle depends. The architectural placement of this layer was described in Chapter 3; the methodological concern here is how raw simulator observations become stable, inspectable variables for later benchmarking, replay, and help-cycle analysis.

4.1.1 Variable Resolution and Source-Missing Tracking

The procedure pack declares a telemetry map that projects raw DCS-BIOS fields onto normalised runtime variables. The resolver produces a flat dictionary of key-value pairs and separately tracks unavailable sources in `vars_source_missing`. This separation is important because a false value and an unobserved value should not have the same methodological meaning. A switch known to be off can support corrective guidance. A missing source supports only uncertainty-aware guidance or fallback.

4.1.2 Gating Rule Evaluation

Each procedure step contains precondition and completion gates. The gate DSL supports threshold checks, range checks, boolean flags, and elapsed-time conditions. Missing, source-missing, or sentinel values fail with diagnostic reasons rather than being silently coerced. The failed gate and reason are included in the help-cycle context and can later be inspected in logs. This turns procedure progress into a rule-governed evidence claim rather than a model-inferred narrative.

4.1.3 Delta Sanitisation and Aggregation

Raw telemetry can change at high frequency because of simulator physics, instrument jitter, and transitional switch states. Delta sanitisation applies per-variable thresholds, debounce windows, and filtering rules. Delta aggregation then compresses recent, sanitised changes into a prompt-safe summary. The resulting deltas are used as evidence of recent learner action without exposing every raw simulator frame to the language model.

4.1.4 Deterministic Fallback

Fallback is part of the method rather than only an implementation safeguard. When model output is unavailable or invalid, the runtime computes a conservative hint from the active step, unsatisfied gates, recent actions, and evidence availability. The fallback distinguishes hard blockers, where observed evidence shows that a gate is unsatisfied, from soft blockers, where required evidence is missing. Overlay guidance is suppressed unless a local rule identifies a unique valid target.

4.2 Knowledge Grounding and Source Policy

Retrieved procedural knowledge supplies textual context for generated help, but it is deliberately source-controlled. The local retrieval adapter indexes curated procedure and aircraft documentation at sentence level and ranks snippets with BM25 [19]. A grounding query is constructed from the active step, its description, and the current observation; the top snippets are included in the help-cycle context.

A declarative source policy restricts retrieval to allowed document sources. This prevents the model from receiving procedure variants outside the active procedure pack, such as carrier-specific instructions when the current profile is an airfield cold start. In

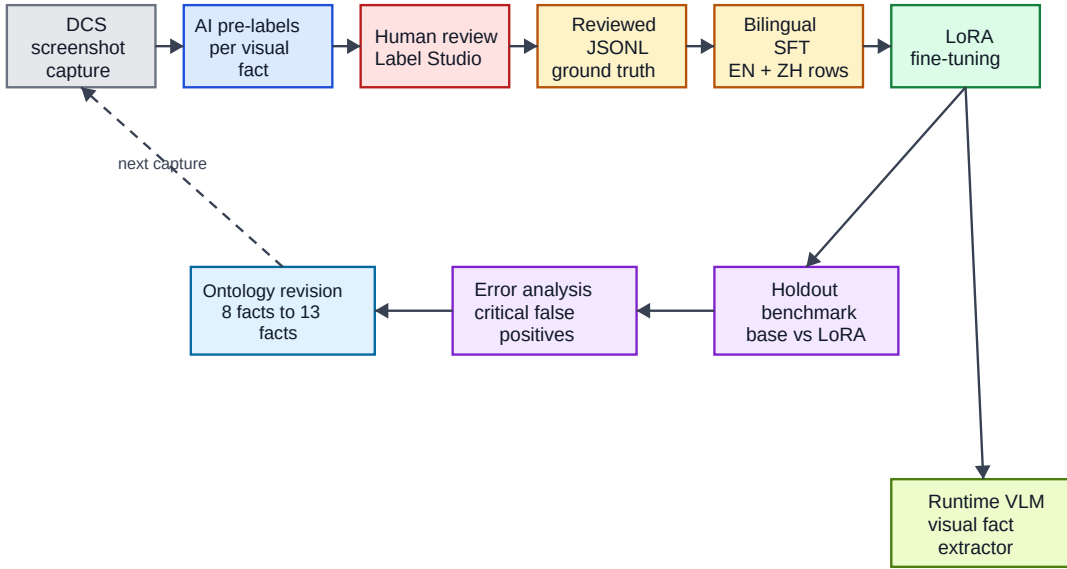
the thesis method, retrieval is therefore not a general document search facility. It is a controlled evidence channel whose scope is part of the procedure configuration.

4.3 Visual Fact Data Pipeline

The visual fact pipeline is the main methodology for RQ2. It turns DCS cockpit screenshots into supervised examples and then into a benchmarked visual fact extractor. The pipeline is designed for traceability: a benchmark error should be traceable back to the model prediction, the human-reviewed label, the pre-label, and the original image.

Figure 4.1 shows the pipeline as a loop. Holdout failures are not only final results; they can reveal weaknesses in the ontology and guide later data collection.

Small-data VLM adaptation pipeline



Each benchmark prediction remains traceable to a reviewed label and original cockpit image.

Figure 4.1: VLM adaptation pipeline. DCS composite screenshots are pre-labelled, human-reviewed, converted into bilingual supervised fine-tuning rows, adapted with LoRA, evaluated on independent holdout sets, and used to drive ontology revision.

1. **Screenshot capture.** DCS renders the left DDI, AMPCD, and right DDI into one composite-panel image. The same composite layout is used for data collection,

pre-labelling, training, benchmarking, and runtime inference. This keeps the input distribution stable across the method.

2. **VLM pre-labelling.** A large external VLM produces initial labels and evidence notes for each image. Pre-labelling is used as an annotation aid, not as ground truth.
3. **Human review.** Pre-labelled samples are reviewed in Label Studio [20]. The reviewer corrects fact states and evidence notes while preserving a link to the raw image and pre-label.
4. **Reviewed JSONL export.** The reviewed export is the authoritative ground-truth artifact for downstream training and evaluation.
5. **Bilingual SFT generation.** Each reviewed image yields an English and a Chinese training row with the same image and fact labels. The assistant target is structured JSON containing visual facts and a short summary; external management fields such as frame identifiers, file paths, and confidence fields are excluded from the target.
6. **LoRA adaptation.** The SFT rows are used to train LoRA adapters for VLM backbones [12]. The implementation uses PEFT adapter definitions [16], TRL supervised fine-tuning infrastructure [22], and Unsloth acceleration for 4-bit VLM training [10].
7. **Holdout benchmark and revision.** Base and LoRA configurations are evaluated on independent holdout sets. Errors are inspected at fact level, especially for completion-critical facts, and the resulting failure modes feed back into ontology and dataset revisions.

The method intentionally keeps the VLM’s task narrow. The model is trained to report visual facts with states `seen`, `not_seen`, or `uncertain`. It is not trained to decide procedure completion or produce tutor actions. Those decisions remain in the procedure engine and harness.

4.4 Dataset Curation

4.4.1 Dataset Overview

Six reviewed datasets support the visual fact work across two ontology generations. Table 4.1 lists their roles and tracked counts.

Table 4.1: Dataset inventory used for visual fact extraction.

Run	Dataset	Samples	Facts	Lang.	Role
Run-001	v1 SFT	180	8	en/zh	v1 training source
Run-002	v1 holdout	50	8	en	v1 holdout
Run-002	v2 newfacts holdout	50	13	en	v2 holdout
Run-003	v2 SFT	220	13	en/zh	v2 training source
Run-004	v2 random holdout	100	13	en	v2 holdout
Run-005	composition rebalance	122	13	en/zh	v2 training supplement

Run-003 and Run-005 contain samples whose free-text summary field is blank (82 and 25 samples respectively). These samples remain valid for the visual fact task because every sample still contains a complete fact array. The supervised target used by the runtime and benchmark is the structured fact state, not the optional summary.

4.4.2 Ontology Evolution: 8-Fact v1 to 13-Fact v2

The first ontology contained eight facts over the composite cockpit panel. It established that a small reviewed dataset could improve structured cockpit fact extraction, but it also exposed a representation problem: some targets mixed visual observation with procedural interpretation. The clearest case was `ins_go`, which asked a single-frame VLM to infer an INS completion state rather than report directly visible evidence.

Figure 4.2 illustrates the methodological correction. Compound procedural targets were decomposed into locally observable visual atoms, leaving temporal and procedural interpretation to the tutor runtime.

The current 13-fact ontology follows three design rules:

1. **Decompose compound targets.** The `ins_go` target is replaced by separate visual atoms for GRND alignment text and INS OK text. The FCS-MC sequence is represented by page, intermediate-result, in-test, and final-GO facts.

Visual fact ontology evolution

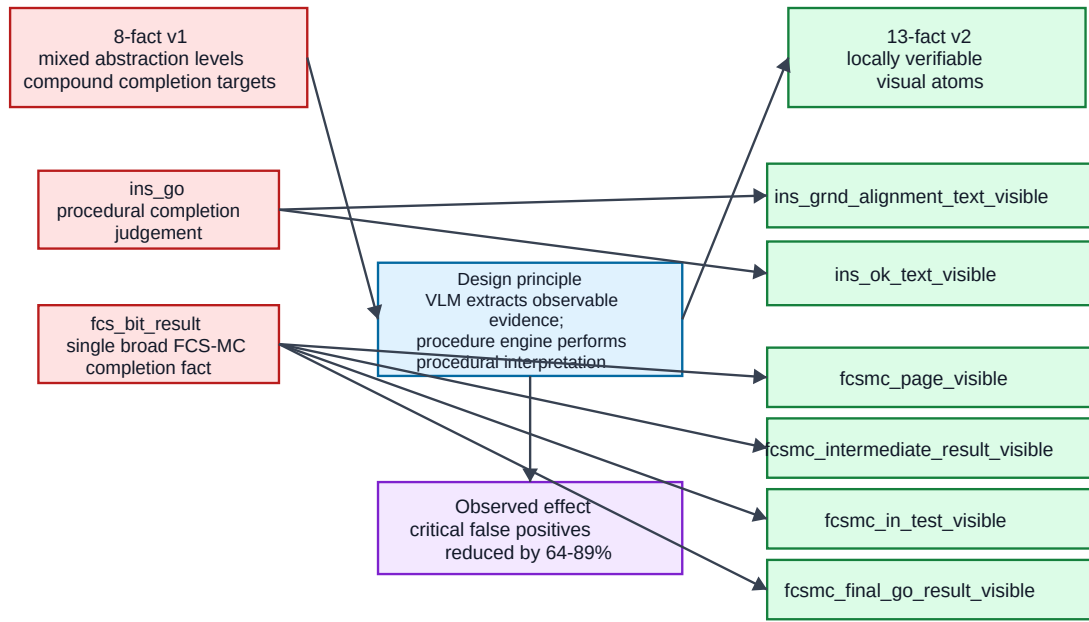


Figure 4.2: Ontology evolution from 8-fact v1 to 13-fact v2. Compound targets such as `ins_go` and broad FCS-MC completion facts were decomposed into locally verifiable visual atoms; the procedure engine then combines those observations with telemetry, phase, and timing evidence.

2. **Separate region from semantics.** Fact names describe the visible state rather than encoding the display region in the identifier. Region constraints are part of the ontology metadata.
3. **Keep facts locally observable.** Each fact should be decidable from the composite image without requiring procedure history. The procedure engine combines facts with telemetry, step bindings, and timing information later.

This ontology discipline is central to the thesis. It defines the boundary between perception and procedural reasoning: the VLM reports visual evidence, while the runtime decides whether that evidence satisfies a procedural step.

4.4.3 Training Recipe: Run-003 + Run-005 $\times$ 2

The primary v2 training recipe combines Run-003 with a repeated Run-005 composition-rebalance supplement. Run-003 contributes 220 reviewed images, exported bilingually as 440 SFT rows. Run-005 contributes 122 reviewed images, exported bilingually as 244 rows; this supplement is included twice, yielding 488 rows. The final recipe therefore contains **928 SFT rows**.

Run-005 targets under-represented positive states from Run-003, including INS OK text, FCS page X marks, and FCS-MC sub-state facts. The repeated supplement is a deliberate composition rebalance strategy, not an independent validation set. For that reason, generalisation claims are based on external holdouts rather than on an internal same-pool evaluation split.

4.4.4 Holdout Independence

Two 13-fact v2 holdouts are used for the main evaluation. Run-002 newfacts contains 50 images from an earlier capture session re-labelled under the 13-fact ontology. Run-004 random contains 100 randomly sampled cockpit-state captures. Local overlap checks report zero exact overlap between either holdout and the Run-003 + Run-005 training union.

The earlier Run-001 evaluation line is treated differently. Because it overlaps with the training data pool, it is useful as a contaminated development benchmark for checking schema learning and historical ontology behaviour, but it is not used as independent evidence for v2 generalisation.

4.5 LoRA Fine-Tuning Configuration

4.5.1 Backbone Models and Comparison Strategy

The primary adaptation line uses `Qwen/Qwen3.5-9B-Base` with the `Run-003 + Run-005×2` recipe. Two supplementary backbones are included to test whether the same data recipe remains meaningful under scale, version, and family changes: `Qwen/Qwen3.6-27B` and `google/gemma-4-31B`. These comparisons are methodological controls rather than separate thesis claims. Chapter 6 reports the numerical results.

The comparison holds the main factors constant wherever possible: ontology, training recipe, benchmark prompt, holdout sets, generation temperature, output schema, and metric computation. Backbone-specific differences remain and are therefore reported explicitly. Qwen-family runs apply LoRA to vision and language modules; the Gemma run uses an all-linear LoRA configuration with vision layers not separately fine-tuned. Full settings appear in Appendix 9, Table 9.1.

4.5.2 Training Stack and Hyperparameters

All backbone runs use 4-bit loading, LoRA rank $r = 16$, $\alpha = 16$, dropout 0.0, four epochs, learning rate 2×10^{-4} , effective batch size 4, maximum sequence length 4096, seed 3407, warmup ratio 0.1, and weight decay 0.01. Adapter weights are saved separately from the base model.

The epoch count and LoRA rank are chosen for small-data domain adaptation. With 928 SFT rows, the adapter must learn both the fixed JSON schema and the visual boundaries of cockpit facts. A very small number of epochs may not teach those boundaries reliably, while excessive training would risk memorising the capture sessions. Rank 16 provides moderate adaptation capacity without turning the adapter into a high-capacity task-specific model.

4.5.3 Bilingual SFT and Composition Rebalance

The bilingual SFT strategy creates two instruction variants per reviewed image: one English and one Chinese. The image and fact labels remain identical. This increases instruction-format coverage without requiring additional image annotation. It does not replace the need for new cockpit states or hard negative examples.

Composition rebalance is handled through Run-005. Because rare positive facts are especially important for downstream procedure reasoning, the training recipe increases

their exposure by adding targeted captures and repeating the Run-005 bilingual rows. The method therefore separates two questions: whether the adapter can learn the ontology from the rebalanced training recipe, and whether the learned extractor generalises to independent holdouts.

4.6 Benchmark Protocol

4.6.1 Metrics

The benchmark compares model predictions against human-reviewed ground truth at both sample and fact level. The metrics are selected to reflect downstream runtime risk:

- `json_valid_rate`: proportion of samples with parseable JSON output.
- `schema_valid_rate`: proportion of samples whose JSON contains the required fact identifiers exactly once and uses only valid states.
- `fact_accuracy`: three-class accuracy over all facts and samples.
- `macro_f1`: macro-averaged F1 over the three states, averaged over facts.
- `seen_f1`: F1 for the positive `seen` class, averaged over facts. This matters because missed or hallucinated visible states change procedure evidence.
- `sample_exact_match`: proportion of samples where all 13 facts match the reviewed labels.
- `critical_false_positive_count`: count of completion-critical facts predicted `seen` when the reviewed label is `not_seen` or `uncertain`.

Critical false positives are tracked separately because they are the most dangerous visual error for the tutor. A false positive on a completion-critical fact could make the procedure engine believe that a display state has been observed when it has not.

4.6.2 Base vs. LoRA Comparison

Base and LoRA configurations receive the same prompt, images, generation parameters, and output schema. Decoding uses temperature 0.0, maximum new tokens 1024, and seed 3407. Predictions are written to JSONL, parsed, normalised, and scored without post-hoc filtering. The comparison artifact reports LoRA-minus- base deltas, while Chapter 6 interprets those deltas.

4.6.3 Benchmark Categories

Benchmark runs are classified by evidence status. Contaminated development sets come from the same data pool as training and are used only for regression and historical interpretation. Held-out new-session benchmarks use independent captures with zero exact training overlap. This distinction determines the strength of the claim that can be made from a metric.

4.6.4 Benchmark Artifacts

Each benchmark run produces a standard directory with metrics for each model, per-sample predictions, error JSONL files, per-fact score CSVs, comparison summaries, human-readable reports, and charts. These artifacts form the evidence chain for the technical results reported in Chapter 6.

4.7 Human-Subject Pilot Methodology

A small formative VR pilot study was conducted to gather descriptive evidence about the tutor in use during the F/A-18C cold-start task. The pilot was not a powered controlled experiment. Its purpose was to exercise the runtime in a realistic VR setting, inspect interaction logs, and identify procedural error patterns for later study design.

4.7.1 Participants and Conditions

Nine participants (P01–P09) completed one trial each. The study used a between-subjects design with non-random assignment based on scheduling availability:

With tutor (n=4). P04, P05, P08, and P09 received overlay highlights, structured help text, telemetry-grounded step inference, and VLM visual fact support where applicable.

Without tutor (n=5). P01, P02, P03, P06, and P07 received no tutor assistance. Telemetry was still recorded for offline step inference and error coding.

Participant experience profiles are reported in Appendix 9, Table 9.5. Because assignment was not randomised and the sample is small, condition comparisons are descriptive only.

4.7.2 Task and Apparatus

All trials used the 33-step F/A-18C cold-start procedure in DCS World on a Meta Quest 3 via Quest Link. The composite cockpit panel was mirrored for experimenter observation. Tutor-condition trials used the live runtime with remote model inference through an OpenAI-compatible endpoint. A trial ended when the participant completed the procedure or voluntarily stopped after being unable to progress.

4.7.3 Data Sources and Coding

The export pipeline produced per-trial summaries, per-step coding files, help-cycle logs for tutor trials, action timelines, quality-gate metadata, and session summaries. Step errors were coded with five categories: omission (OM), commission (CO), order error (OR), prompting/assistance error (PA), and state violation (SV). The automated coding output was reviewed manually using telemetry, help-cycle logs, and action timelines; the human-coded columns are the authoritative source for reported pilot results.

4.7.4 Data-Quality Notes

Two trial-level caveats affect interpretation. First, P08's automatic summary reported incomplete progress, but manual telemetry review confirmed completion of all 33 steps. The manual coding is used for analysis, and the pre-review file is preserved for audit. Second, P02's questionnaire metadata contains an internal participant-id mismatch. P02's task metrics are retained, but P02 is excluded from questionnaire-derived workload or confidence summaries.

4.7.5 Analysis Approach

The pilot analysis reports per-participant outcomes and condition-level descriptive summaries. No null-hypothesis significance tests are performed. Observed durations are documented but not interpreted as completion-time comparisons because incomplete without-tutor trials ended at different stopping points. The analysis focuses on completion status, manual error categories, and tutor interaction patterns such as help requests, VLM calls, overlay actions, and fallback events.

4.8 Summary

The methodology establishes the evidence chain used by the rest of the thesis. Telemetry and retrieval provide controlled runtime context. The visual fact pipeline produces a reviewed and benchmarked perception layer for RQ2. The 13-fact ontology separates perception from procedural reasoning. The LoRA protocol uses matched data and holdouts to evaluate small-data adaptation without relying on contaminated development sets. The pilot protocol provides formative VR evidence for RQ3 while keeping its claims descriptive. Together, these methods make the later results auditable without overstating what the available evidence can prove.

5 Implementation

This chapter covers implementation-level specifics that directly support the research questions, focusing on the evidence-grounding enforcement mechanism, the deterministic fallback path, the VLM runtime integration, and the replay infrastructure. Implementation details not essential to the core thesis argument—CLI command reference, complete technology stack versions, Lua script inventory, and full hyperparameter configuration—are documented in [Appendix 9](#).

5.1 Technology Stack

The Python codebase uses Python 3.10+ with Poetry for dependency management. Core library dependencies are minimal: `httpx` for HTTP, `PyYAML` for declarative configuration, `jsonschema` (Draft 2020-12) for validation of model outputs and telemetry frames, and `Pillow` for image handling in the VLM pipeline. Testing uses `pytest` with branch coverage.

Model serving is provider-agnostic. The primary path uses an OpenAI-compatible API endpoint (configured for vLLM, llama.cpp, or TGI servers). An Ollama fallback provides local compatibility. A deterministic stub returns fixed responses for testing. Multimodal support is toggled via configuration.

The DCS Lua-side integration comprises scripts for state export (DCS-BIOS protocol), overlay rendering (highlight, clear, and pulse commands via DCS’s cockpit highlight API), and composite-panel screenshot capture. LoRA fine-tuning uses Unsloth with PEFT and TRL; pre-labeling and human review are conducted in Label Studio.

5.2 Evidence Grounding Enforcement

The evidence-grounding mechanism described in [Section 3.3](#) is implemented through three coordinated components.

Dynamic JSON Schema generation. The function `build_help_response_schema` generates the HelpResponse JSON Schema at runtime from the current procedure pack’s step identifiers, UI map targets, and error taxonomy codes. For every overlay target in the schema’s `targets` array, a conditional `if/then` sub-schema requires the `evidence` array to contain at least one item referencing that target. Each evidence item must declare a `type` from the five grounding kinds (`var`, `gate`, `rag`, `delta`, `visual`) and carry a non-empty `ref` matching the corresponding prefix family. The schema is cached keyed on configuration file modification times, avoiding redundant regeneration during long-running sessions.

Response validation and mapping. Model output passes through a narrow JSON extraction pipeline (code fence removal and delimiter-based parsing only; transformations outside this set are rejected), schema validation via `Draft202012Validator`, and allowlist checks on step identifiers and overlay targets. The mapping function validates that every overlay target cited in the response has at least one evidence reference whose prefix matches a key actually present in the help-cycle context. Targets without grounding evidence are rejected, and the rejection is recorded in the response metadata.

Safety constraints. The overlay action executor enforces three rules before dispatching to DCS: only `overlay` actions are executable (no control injection or key emulation); every target is verified against the UI target allowlist; at most one target may be highlighted concurrently. When a new highlight is requested, any prior highlight is automatically cleared.

5.3 Deterministic Fallback

The fallback mechanism operates when no model is available, model output fails JSON extraction or schema validation, or response mapping produces an empty result. It examines the currently active step, evaluates unsatisfied gating rules, identifies missing conditions, and classifies each missing evidence source as a hard blocker (gate explicitly fails) or soft blocker (telemetry source absent). This distinction prevents the tutor from incorrectly claiming a step is unsatisfied when evidence is simply unavailable. The fallback produces a text-only hint restricted to the active step; overlay highlights are suppressed unless a local rule can uniquely determine a valid target from the current telemetry state.

5.4 VLM Visual Fact Extraction Runtime

The VLM runtime operates as a pipeline triggered when the active step is vision-priority (S08, S15, S18). A capture request is dispatched to DCS via UDP; the Lua-side integration renders the three primary cockpit displays into a single composite PNG image. The Python-side buffered vision session maintains a sliding window of recent observations and selects both a pre-trigger frame (most recent frame before the trigger, within a 250 ms sync window) and the trigger frame.

The selected frames are base64-encoded and inserted alongside the fact extraction prompt into a multimodal chat-completion request. The prompt enumerates all 13 facts, describes the three-region composite-panel layout, and provides detailed decision-boundary instructions per fact. The raw model output is parsed, sanitised (unknown keys and invalid fact IDs rejected, states restricted to the three admissible values), and validated against a dynamically generated JSON Schema. Sanitised facts are merged into the runtime vision fact snapshot with sticky preservation and TTL-based expiry.

When the VLM is unavailable or the multimodal request fails, the help cycle proceeds with telemetry and knowledge only—a deliberate degradation property rather than a system failure.

5.5 Replay and Logging Infrastructure

The JSONL event store is the sole persistence mechanism for runtime events, implementing append-only writing with immediate flush after each write for crash-safe logging. The replay module provides two entry points: full simulation using a mock environment adapter that replays pre-recorded observation sequences, and log validation that reconstructs the step state machine and verifies legal state transitions against a procedure pack.

The replay evaluation infrastructure supports suite-driven regression replay: a YAML suite file specifies test cases with expected outcomes, model configuration, and optional vision frame directories. Five replay cases cover distinct scenario types (no-operation, battery-on, engine-off, FCS BIT, INS alignment), each with DCS-BIOS raw captures and per-frame sidecar files. The suite also includes 20 live-fixture regression entries that validate specific evidence-grounding and overlay-target correctness properties (Section 3.6).

Interaction metrics are derived from event logs: stall time per step, help request count,

LLM trigger count, UI click and failure counts, and highlight request and acknowledgment counts.

5.6 Live Runtime Orchestration

The live runtime orchestrates a continuous loop: DCS-BIOS UDP frames are ingested and enriched into normalised state variables; when a help trigger is received (hotkey or UDP message), the procedure engine computes a deterministic step hint, a vision capture is triggered if the active step requires it, BM25 knowledge snippets are retrieved, the full prompt is assembled and sent to the model adapter, the response is validated and mapped to overlay actions, and the complete help cycle is logged. The runtime supports both real-time DCS operation and replay of pre-recorded BIOS captures through a shared code path.

Model access is configured through environment variables with a unified naming convention; the runtime validates all required configuration at startup and redacts API keys from all log output.

6 Evaluation

This chapter presents empirical evidence organised around the three research questions defined in Chapter 1. Section 6.1 (RQ2) reports the VLM technical benchmark results on the 13-fact ontology across three backbone models. Section 6.2 (RQ1) assesses system-level technical readiness through replay evaluation and harness regression testing. Section 6.3 (RQ3) reports the formative VR pilot study with nine participants.

Figure 6.1 gives the chapter-level evidence map. The three evaluation lines use different artifacts and support different kinds of claim: benchmark metrics for RQ2, regression and replay evidence for RQ1, and descriptive pilot data for RQ3.

6.1 RQ2: VLM Technical Evaluation

The technical evaluation addresses whether a LoRA-fine-tuned VLM can reliably extract structured visual facts from F/A-18C cockpit screenshots on holdout data the model has never seen during training. All metrics are computed by the automated benchmark harness described in Section 4.6.

6.1.1 Dataset and Training Summary

The visual fact extraction task evolved across two generations of fact ontology. The first generation (v1) defined 8 visual facts and established the initial training pipeline. The second generation (v2) expanded the ontology to 13 facts, aligned with the runtime contract. Table 6.1 summarises the six annotated datasets.

The v2 training recipe combines Run-003 (220 reviewed tasks) with Run-005 (122 reviewed tasks, collected to rebalance under-represented fact states), with Run-005 applied twice ($\text{Run-005} \times 2$). Bilingual export (EN + ZH) yields a combined training set of **928 rows**. The holdout sets—Run-002 newfacts (50 samples) and Run-004 random (100 samples)—are English-only with verified **zero exact overlap** with training data.

Evaluation data flow across research questions

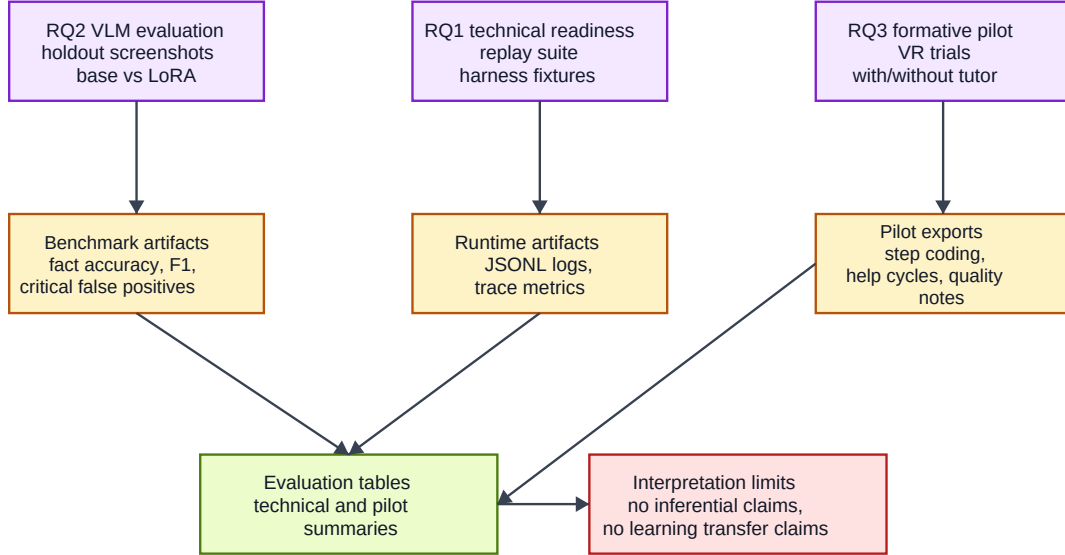


Figure 6.1: Evaluation and data flow across research questions. VLM benchmark artifacts support RQ2, replay and harness logs support RQ1 technical readiness, and VR pilot exports support the descriptive RQ3 findings.

Table 6.1: Summary of annotated visual fact datasets.

Dataset	Run	Facts	Tasks	Bilingual	Role
vision_sft	Run-001	8	180	yes	v1 train
vision_sft_holdout_run002	Run-002	8	50	no	v1 hold
vision_sft_holdout_run002_newfacts	Run-002	13	50	no	v2 hold
vision_sft_run003	Run-003	13	220	yes	v2 training
vision_sft_run005_composition_rebalance	Run-005	13	122	yes	v2 rebal
vision_sft_holdout_run004_random	Run-004	13	100	no	v2 hold

6.1.2 Qwen3.5 13-Fact V2 Results

The primary technical result is the performance of the Qwen3.5-9B LoRA adapter trained on Run-003 + Run-005 $\times$ 2 (928 rows, 4 epochs, learning rate 2×10^{-4} , LoRA $r = 16$, $\alpha = 16$). Table 6.2 reports results on two independent holdout sets.

Table 6.2: 13-fact v2 benchmark: Qwen3.5-9B Base vs. LoRA on two holdout sets.

Metric	Run-002 newfacts (50)			Run-004 random (100)		
	Base	LoRA	Δ	Base	LoRA	Δ
JSON valid	1.0	1.0	—	0.96	1.0	+0.04
Schema valid	1.0	1.0	—	0.96	1.0	+0.04
Fact accuracy	0.8677	0.9908	+0.1231	0.8038	0.9931	+0.1892
Macro F_1	0.4513	0.6515	+0.2002	0.4393	0.6100	+0.1707
Seen F_1	0.5022	0.9648	+0.4626	0.5659	0.9159	+0.3500
Exact match	0.10	0.88	+0.78	0.03	0.92	+0.89
Critical FP	11	4	−7	70	8	−62

The LoRA adapter achieves fact accuracy of 0.9908 (Run-002) and 0.9931 (Run-004), with sample exact match rates of 0.88 and 0.92 respectively. Critical false positives drop by 64% (11 $\rightarrow$ 4) on Run-002 and 89% (70 $\rightarrow$ 8) on Run-004. The base model degrades markedly on the larger Run-004 holdout (4 schema-invalid responses, 70 critical false positives concentrated on `fcsmc_intermediate_result_visible`), while the LoRA adapter restores format reliability and suppresses indiscriminate hallucination.

The residual critical false positives in the LoRA adapter are concentrated on completion-type facts: `fcsmc_final_go_result_visible` (8 on Run-004, 1 on Run-002) and `ins_ok_text_visible` (2 on Run-002).

6.1.3 Gemma-4 and Qwen3.6 Backbone Comparison

To assess whether the training recipe generalises, the identical data recipe (Run-003 + Run-005 $\times$ 2, 928 rows, 13-fact ontology) was applied to two additional backbones. Table 6.3 summarises the comparison.

Qwen3.6-27B’s base model starts substantially stronger than Qwen3.5-9B (fact accuracy 0.91 vs. 0.87 on Run-002) but the LoRA gain is narrower ($\Delta = +0.077$ vs. +0.123), consistent with the expectation that larger models have less adaptation headroom under small-data fine-tuning. The Qwen3.6 LoRA adapter achieves comparable performance to Qwen3.5 on most metrics but does not surpass the 9B line.

Table 6.3: 13-fact v2 results across three backbones (LoRA models only).

Metric	Run-002 newfacts (50)			Run-004 random (100)		
	Q3.5-9B	Q3.6-27B	G-4-31B	Q3.5-9B	Q3.6-27B	G-4-31B
Fact accuracy	0.9908	0.9877	0.9492	0.9931	0.9892	0.9715
Seen F_1	0.9648	0.9479	0.8123	0.9159	0.9147	0.7959
Exact match	0.88	0.86	0.44	0.92	0.86	0.74
Critical FP	4	3	0	8	8	13

Gemma-4-31B achieves zero critical false positives on Run-002, reflecting a more conservative error style, but at the cost of lower recall (seen F_1 0.81 vs. 0.96 for Qwen3.5 on Run-002) and lower sample exact match (0.44 vs. 0.88). This backbone-level difference in error style—conservative (Gemma) vs. assertive (Qwen)—is relevant for safety-critical deployment decisions.

Overall, Qwen3.5-9B remains the best-performing configuration on fact accuracy, seen F_1 , and sample exact match across both holdouts. The data recipe transfers across all three backbones with consistent improvements over their respective base models.

6.1.4 Error Analysis

The 13 facts can be grouped into three difficulty categories based on the Qwen LoRA adapter’s performance. **Coarse panel presence** (6 facts: `tac_page_visible`, `supt_page_visible`, `fcs_page_visible`, `bit_root_page_visible`, `fcsmc_page_visible`, `hsi_page_visible`): near-perfect classification, with the single exception of two false negatives on `tac_page_visible` on Run-002. **Fine-grained state indicators** (5 facts: `fcs_page_x_marks_visible`, `fcsmc_intermediate_result_visible`, `fcsmc_in_test_visible`, `fcsmc_final_go_result_visible`, `hsi_map_layer_visible`): perfect on four facts; residual false positives on `fcsmc_final_go_result_visible` (8 on Run-004). **Small-text recognition** (2 facts: `ins_grnd_alignment_text_visible`, `ins_ok_text_visible`): strong on the alignment text (one false positive on Run-002); 2 false positives on `ins_ok_text_visible` on Run-002.

The base model’s dominant failure mode is unidirectional hallucination on rare states: `fcsmc_intermediate_result_visible` appears in only 7 of 100 Run-004 samples, yet the base model asserts it as **seen** in 60 additional samples. The LoRA adapter eliminates this class of error almost entirely, shifting the error profile from diffuse hallucination to focused, interpretable residual errors on completion-type facts.

6.2 RQ1: System-Level Technical Readiness

Beyond the component-level VLM benchmarks, the repository contains evidence of systematic validation at the system level.

6.2.1 Replay Evaluation

The replay evaluation suite at `replay_eval/fa18c_startup_v04/` defines five test cases exercising distinct scenario types: no-operation, battery-on, battery-and-generator-off, FCS reset and BIT, and INS alignment (see Appendix 9, Table 9.4). Each case includes DCS-BIOS raw captures and, where applicable, per-frame sidecar files for vision-dependent steps. All five cases produce the expected step inference and overlay targets, confirming that the telemetry pipeline, procedure engine, and vision path function correctly end-to-end when driven by recorded data.

6.2.2 Harness Fixture Regression

The step harness (Section 3.6) is covered by 20 live-fixture regression entries in the coverage matrix, each replaying a specific telemetry snapshot through the full help cycle and asserting a specific expected outcome. The fixtures cover: correct advancement on completed conditions, safe rejection of wrong-target proposals (e.g., PB18 repaired to PB5 for BIT root sub-state, S10 left-engine guidance using left-click rather than right-click semantics), prevention of VLM evidence leakage into non-visual steps, correct handling of moving/settling controls (refuelling probe, arresting hook), consistent interaction semantics (mouse wheel for rotary controls), and correct repair of partially completed steps.

6.2.3 Experiment Export Infrastructure

The experiment export pipeline (Section 4.7) has been exercised on nine live VR trials, producing complete artifact sets (`trial_summary`, `step_coding`, `help_cycles`, `action_timeline`, `quality_gate`) for each. Quality gates passed for all nine trials, with configuration hashes verified consistent across the study. This demonstrates that the instrumentation pipeline is functional for human-subject data collection, though the participant/session metadata schema remains incomplete (missing questionnaire linkage fields noted in P04's quality gate warnings).

6.3 RQ3: Formative VR Pilot Study

A small formative pilot study was conducted with nine participants to gather descriptive evidence on how the tutoring system supports learners during the F/A-18C cold-start procedure compared with unaided practice. The study should be interpreted as a preliminary field trial, not a powered controlled experiment. Methodology details are reported in Section 4.7.

6.3.1 Participant and Trial Overview

Table 6.4 summarises each participant’s condition, completion status, step completion, observed duration, and key interaction counts. All data are drawn from manual step coding (`step_coding.csv`) and trial summaries (`trial_summary.csv`).

Table 6.4: Participant and trial overview. Observed duration is the wall-clock interval from first telemetry event to last; it is not a completion time measure for incomplete trials.

P	Condition	Completed	Steps	Dur. (s)	Help	VLM	Ovrl.	Fallbk.
P01	without_tutor	no	23/33	2015	0	0	0	0
P02	without_tutor	no	27/33	739	0	0	0	0
P03	without_tutor	no	24/33	243	0	0	0	0
P06	without_tutor	no	29/33	269	0	0	0	0
P07	without_tutor	no	19/33	218	0	0	0	0
P04	with_tutor	yes	33/33	442	19	6	22	6
P05	with_tutor	yes	33/33	317	7	2	6	2
P08	with_tutor	yes*	33/33	332	7	5	10	4
P09	with_tutor	yes	33/33	1033	44	10	44	9

*P08: auto-summary reports incomplete; manual coding confirms 33/33 completed (see data-quality note below).

Completion. All four with-tutor participants completed the full 33-step procedure. None of the five without-tutor participants completed the full procedure; they stopped after completing 19–29 steps. The most commonly missed steps in the without-tutor condition were S18 (FCS-MC BIT page, missed by 5/5 participants), S19 (FCS BIT final GO, missed by 5/5), and S27 (flap switch to AUTO, missed by 5/5).

Observed duration. With-tutor trials ranged from 317 s to 1033 s (median 387 s). Without-tutor trials ranged from 218 s to 2015 s. Because without-tutor trials ended at different procedure points when participants could not progress, these durations cannot be compared as completion times.

6.3.2 Procedural Errors

Table 6.5 reports manual error counts per participant.

Table 6.5: Manual procedural error counts per participant (from manual step coding).
OM = omission, CO = commission, OR = order error, PA = prompting/ assistance error, SV = state violation.

P	Condition	OM	CO	OR	PA	SV
P01	without_tutor	8	0	1	0	1
P02	without_tutor	0	0	0	1	0
P03	without_tutor	7	0	2	0	2
P06	without_tutor	2	0	0	2	0
P07	without_tutor	11	0	0	0	0
P04	with_tutor	0	3	0	3	1
P05	with_tutor	0	3	0	3	0
P08	with_tutor	0	0	0	0	0
P09	with_tutor	0	3	0	2	0

Omission errors. The without-tutor condition shows a clear omission pattern: four of five participants recorded 7–11 OM errors (missed steps). In contrast, the with-tutor condition recorded zero OM errors—all 33 steps were completed by all four participants.

Commission and prompting errors. The with-tutor condition shows a different error profile: CO errors (3 per participant in P04, P05, P09) and PA errors (2–3 per participant) replace the omission errors observed in the without-tutor condition. These are primarily assistance-coupled errors—cases where the tutor provided guidance that was followed but did not fully align with the procedure specification, or where the participant executed a step correctly in substance but with minor deviation from the prescribed sequence.

Error totals. Without-tutor participants averaged 7.4 manual errors; with-tutor participants averaged 4.5. However, the error categories differ qualitatively: without-tutor errors are dominated by omissions (mean OM = 5.6), while with-tutor errors are exclusively CO and PA (mean CO = 2.3, mean PA = 2.0).

6.3.3 Tutor Interaction

Table 6.6 summarises tutor interaction metrics for the four with-tutor trials.

All help cycles used the model generation path (`generation_mode = model`); no cycles fell back to deterministic-only mode. VLM calls were issued for vision-priority

Table 6.6: Tutor interaction summary (with-tutor condition only).

P	Help cycles	VLM calls	Overlay exec.	Overlay rej.	Fallback
P04	19	6	22	0	6
P05	7	2	6	0	2
P08	7	5	10	0	4
P09	44	10	44	0	9
Total	77	23	82	0	21

steps (S08, S15, S18) and totalled 23 across four trials. **Overlay rejection rate was zero:** all 82 overlay actions proposed by the tutor passed evidence-grounding, allowlist, and schema validation. The 21 fallback events ($21/77 = 27\%$ of help cycles) occurred when the model’s proposed overlay targets were repaired by the harness—for example, when the model proposed a target that the harness re-mapped to a more specific sub-state target (e.g., PB18 repaired to PB5 for BIT root).

6.3.4 Data-Quality Notes

1. **P08 auto-summary correction.** The automated pipeline reported P08 as incomplete (5/33 steps). Manual telemetry review confirmed all 33 steps reached required states. The auto-pipeline failed to reconstruct completion evidence for vision-priority steps from passive telemetry. Manual coding overrides the auto-summary; P08 is reported as completed with 0 manual errors.
2. **P02 questionnaire exclusion.** P02’s questionnaire JSON file contains internal `participant_id = P01`, indicating mislabelling. P02 is excluded from any workload, TLX, or confidence summary. Task metrics (steps, errors) from `trial_summary.csv` and `step_coding.csv` are used.
3. **Duration interpretation.** Without-tutor trials ended at different procedure points when participants could not progress further. Observed durations are reported for completeness but do not support completion-time comparison between conditions.
4. **Non-random assignment.** Participants were assigned to conditions based on scheduling availability, not randomisation. Prior DCS and VR experience varied across participants (see Appendix 9, Table 9.5).

6.3.5 Summary of Pilot Findings

The pilot provides three pieces of formative evidence. First, all four tutor-condition participants completed the full 33-step procedure, whereas none of the five without-tutor participants did, suggesting the tutor enables procedural progression for these participants under these conditions. Second, the tutor condition eliminated omission errors in the coded data, but did not eliminate all errors: assistance-coupled CO and PA errors appeared in three of four with-tutor trials. Third, the tutoring infrastructure—telemetry grounding, overlay dispatch, evidence validation—operated without overlay rejections across 77 help cycles, supporting the technical feasibility of evidence-grounded tutoring in a live VR setting.

These findings are descriptive and formative. They do not establish that the tutor improves learning, reduces workload, or accelerates skill acquisition relative to alternative methods. The pilot’s small sample ($n = 9$), non-random assignment, and variable participant experience preclude inferential conclusions. Larger controlled evaluation addressing these limitations is required.

7 Discussion

This chapter interprets the results presented in Chapter 6 and situates them within the larger trajectory of the thesis. Section 7.1 explains why the project evolved from the original VR-first proposal toward the telemetry- and VLM-heavy implementation reported here, and how the formative pilot study fits within this trajectory. Section 7.2 interprets what the benchmark results do and do not demonstrate. Section 7.3 interprets the formative VR pilot findings. Section 7.4 draws design lessons from the two-generation evolution of the visual fact ontology. Section 7.5 enumerates the known limitations of the current system and experimental evidence. Section 7.6 outlines directions for future work that follow from these limitations.

7.1 Evolution from Proposal to Implemented System

The original thesis proposal [24] specified a VR-first grounded tutoring system with gaze, hand-tracking, and voice input on the Meta Quest 3, supported by a lightweight RAG+LLM backend and evaluated through a three-condition human-subject study. The work reported in this thesis differs from that vision in two substantive ways.

First, the implementation effort shifted from VR-native interaction to telemetry-grounded tutoring on desktop DCS, because the core technical challenges—reliable telemetry processing, evidence grounding, VLM-based visual fact extraction, and harness-level output validation—required a stable development and debugging environment that desktop DCS provides. This shift produced the ports-and-adapters architecture, the procedure pack and gating engine, the deterministic fallback path, the JSONL logging and replay infrastructure, and the VLM adaptation pipeline—none of which were anticipated in the original proposal.

Second, the VLM adaptation work grew from an unplanned component into one of the thesis’s main contributions. The original proposal assumed read-only simulator state flags would suffice for procedural state tracking; in practice, several critical procedure steps require visual confirmation of display-page content not exported by DCS-BIOS. Ad-

addressing this gap led to the visual fact ontology, the seven-stage data pipeline, the LoRA fine-tuning experiments, and the three-backbone benchmark comparison—forming a self-contained contribution with verified holdout evidence.

The formative VR pilot study (RQ3) represents a partial convergence back toward the original proposal’s human-subject evaluation ambition. The pilot used the Quest 3 with the full tutoring runtime, produced real participant data, and demonstrated that the evidence-grounded infrastructure operates correctly in a live VR setting. However, the pilot was small ($n = 9$), non-randomised, and descriptive rather than inferential. It does not fulfil the proposal’s three-condition controlled evaluation goal. The gap between the formative pilot and a fully powered controlled study remains the central unfinished thread of the thesis.

7.2 Interpretation of Technical Results

The benchmark results reported in Section 6.1 provide evidence about the performance of the VLM visual fact extraction pipeline. This section draws interpretive conclusions from those results, distinguishing carefully between what the benchmarks demonstrate and what they do not.

7.2.1 What the Benchmark Results Demonstrate

Small-data LoRA adaptation greatly improves domain-specific VLM extraction. Under the 13-fact v2 ontology, the Qwen3.5-9B LoRA adapter trained on 928 rows (Run-003 + Run-005 $\times$ 2) achieves a fact accuracy of 0.9908 on the Run-002 newfacts holdout and 0.9931 on the Run-004 random holdout, compared to 0.8677 and 0.8038 for the base model, respectively. The seen F_1 rises from 0.50–0.57 (base) to 0.92–0.96 (LoRA). These are large effect sizes by any standard, and they are observed on holdout sets with verified zero training overlap, ruling out memorisation as an explanation.

The data recipe transfers across model backbones. The identical training recipe (Run-003 + Run-005 $\times$ 2, bilingual, 13-fact ontology) applied to the Gemma-4-31B backbone produces a similar pattern of consistent improvement: fact accuracy rises from 0.82 to 0.95 on Run-002 and from 0.81 to 0.97 on Run-004. This cross-backbone transfer provides evidence that the improvement is driven by the data and ontology design rather than by idiosyncratic properties of a single model architecture.

JSON and schema validity become near-perfect after training. On the Run-

004 random holdout, the base model produces 4 schema-invalid or JSON-invalid responses out of 100; the LoRA adapter produces none. This is not a trivial property: in a runtime system that must parse model output to extract overlay targets and step recommendations, format reliability is a hard requirement. The fine-tuning process reliably teaches the model to conform to the expected output schema.

The error profile shifts from indiscriminate hallucination to targeted false positives on completion-type facts. The base model’s dominant failure mode is unidirectional hallucination: it asserts `seen` for many facts that are not present, particularly on rare states such as `fcsmc_intermediate_result_visible` (60 false positives on Run-004 out of only 7 true occurrences). The LoRA adapter eliminates this class of error almost entirely, concentrating its residual errors on a specific subset of facts: `fcsmc_final_go_result_visible` (8 false positives on Run-004) and, to a lesser extent, `ins_ok_text_visible` (2 false positives on Run-002). This shift from diffuse hallucination to focused, interpretable error patterns is a qualitative improvement that is not fully captured by aggregate accuracy metrics.

The benchmark protocol itself constitutes a contribution. The use of holdout sets with verified zero overlap, per-fact confusion analysis, critical false positive tracking, and automated chart generation provides a reproducible evaluation framework that can be reused for future iterations of the ontology, for different backbone models, and for other procedures.

7.2.2 What the Benchmark Results Do NOT Demonstrate

They do not demonstrate that the tutoring system improves learning. The benchmarks measure the accuracy of visual fact extraction from cockpit screenshots. Accurate fact extraction is a necessary condition for a grounded tutoring system to function correctly, but it is not a sufficient condition for improving human learning outcomes. A learner could receive perfectly accurate visual facts and still fail to learn the procedure, or could learn effectively from a system with imperfect fact extraction. The causal chain from perception accuracy to learning outcomes has not been tested.

They do not demonstrate that visual fact extraction leads to better tutoring. The benchmarks evaluate the VLM component in isolation, not the end-to-end tutoring system. Whether the availability of visual facts improves the quality, timeliness, or acceptability of the tutor’s help responses relative to a telemetry-only baseline is an open question that requires a system-level evaluation.

They measure a perception subtask, not pedagogical effectiveness. The

metrics reported — fact accuracy, seen F_1 , sample exact match, critical false positives — are perception metrics. They answer the question “does the VLM see the cockpit correctly?” rather than “does the tutoring system help the learner?”. Conflating the two would be a category error.

7.2.3 The `ins_go` to `ins_ok_text` Decomposition as a Design Lesson

The 8-fact v1 ontology included a single fact, `ins_go`, which asked the VLM to recognise whether the INS alignment had reached a visible GO state. This fact proved to be the dominant source of false positives in the v1 adapter: 13 of 50 Run-002 samples were incorrectly predicted as `seen`. The issue was not that the VLM failed to learn from the training data, but that the target itself was poorly specified. `ins_go` was not a visual fact in the same sense as “the FCS page is visible on the left DDI”; it was a procedural completion judgement that required integrating the presence of specific text (“OK”), the absence of other text (alignment countdown), and the procedural context (which phase of the alignment the system is in). Asking a single-frame VLM to make this judgement conflated perception with reasoning.

The 13-fact v2 ontology decomposes this compound target into lower-level, visually observable facts: `ins_grnd_alignment_text_visible` and `ins_ok_text_visible` each describe a specific piece of text on the AMPCD. The procedural interpretation — whether the combination of these facts indicates that alignment is complete — is deferred to the downstream procedure engine, which has access to telemetry, procedure phase, and timing information. This decomposition was not a cosmetic change; it was the single most important design decision in the ontology evolution, and the benchmark results bear out its effectiveness: `ins_grnd_alignment_text_visible` achieves a seen F_1 of 0.99 on Run-002, and `ins_ok_text_visible` produces only 2 false positives on Run-002 and zero on Run-004.

7.3 Interpretation of Pilot Findings

The formative VR pilot study (Section 6.3) provides descriptive evidence that must be interpreted within its proper scope.

Completion and omission errors. The clearest signal is the completion gap: 4/4 with-tutor participants completed all 33 steps, while 0/5 without-tutor participants did.

Coupled with the error coding—zero OM errors in the tutor condition versus a mean of 5.6 OM errors in the without-tutor condition—this suggests the tutor helped participants progress through the procedure by reducing missed steps. This is consistent with the tutor’s design: the help system actively identifies the next required step and highlights the relevant cockpit control, making it harder for a participant to skip a step unknowingly.

Assistance-coupled errors. The tutor condition did not eliminate errors; it changed their composition. CO and PA errors appeared in three of four with-tutor trials, with a total of 9 CO and 8 PA errors across the condition. These are primarily assistance-coupled: cases where the tutor provided guidance that was followed but where the resulting action did not fully align with the procedure specification. For example, the harness occasionally repaired model-proposed overlay targets (PB18 to PB5 for BIT root sub-state), and these repairs, while correct, could produce a participant action that was technically the right control but in a slightly different context than the strict procedure definition. These errors highlight the inherent tension between flexible LLM-generated guidance and fixed procedural specifications: the tutor can guide correctly in substance while deviating in form.

Infrastructure reliability. The zero overlay rejection rate (0/82) across 77 help cycles provides evidence that the evidence-grounding, allowlist, and schema validation pipeline functions correctly in live VR operation. All 21 fallback events were harness repairs (target re-mapping), not rejections. This supports the architectural claim that evidence grounding can be enforced at the schema level without blocking legitimate tutor output.

What the pilot does not show. The pilot provides no evidence about learning (no pre/post knowledge measures), retention (no follow-up testing), workload (P02 questionnaire data unresolved; other questionnaire data not yet fully coded), or transfer. The small sample, non-random assignment, and variable participant experience mean that observed differences between conditions cannot be attributed to the tutor alone. The pilot’s value is formative: it demonstrates that the instrumentation pipeline captures the intended data, that the tutor functions end-to-end in VR, and that the experimental design is executable.

7.4 Ontology Design Lessons

The two-generation evolution from 8-fact v1 to 13-fact v2 provides concrete design lessons that may inform future visual ontology construction for procedural training systems.

7.4.1 Abstraction-Level Discipline

The central lesson is that a VLM-based visual fact extractor should be asked to report *observable visual evidence*, not *procedural conclusions*. The 8-fact v1 ontology mixed these two levels of abstraction within a single fact set: facts such as `fcs_page_visible` and `bit_root_page_visible` described visually localisable display-page states, while facts such as `ins_go` and `fcs_bit_result_visible` encoded higher-level procedural judgements. The mixed-abstraction ontology produced the predictable consequence: straightforward page-visibility facts achieved strong performance, while the compound procedural facts accounted for the majority of critical false positives.

The 13-fact v2 ontology enforces a design discipline in which every fact is a locally verifiable piece of visual evidence. This discipline has three practical consequences. First, it makes the per-fact annotation task clearer for human reviewers: deciding whether “OK” text is visible on the AMPCD is a well-defined visual judgement; deciding whether “INS alignment is complete” is not. Second, it produces cleaner training signals, because the mapping from image content to fact label is more direct and less dependent on implicit procedural context. Third, it localises errors: when a fact is misclassified, the source of the error can be inspected in the image region that the fact describes, without needing to reason about the model’s implicit understanding of procedure phase.

General design principle. The experience of moving from a mixed-abstraction ontology (v1) to a disciplined one (v2) yields a design principle that is explicit and transferable:

Design Principle: VLM fact targets should be low-level, locally verifiable visual propositions. Higher-level procedural state judgments should be synthesised by downstream reasoning from multiple visual facts combined with temporal and telemetry context.

This principle formalises the boundary between perception and reasoning. The VLM is responsible for answering “what text, symbol, or display element is visible in this image region?”—questions that have ground-truth answers determinable from a single frame. The procedure engine is responsible for answering “what does this combination of visual

facts, telemetry variables, and temporal context imply about the current procedure state?”—questions that require multi-source integration. Enforcing this boundary at the ontology design stage, rather than discovering it through error analysis after training, is the primary design lesson of the ontology evolution.

7.4.2 Decomposition of Compound Targets

The effectiveness of this decomposition is measurable. On the Run-002 holdout, the 8-fact v1 LoRA adapter produced 15 critical false positives, of which 13 (87%) were on the compound target `ins_go`. After decomposing `ins_go` into the low-level visual propositions `ins_grnd_alignment_text_visible` and `ins_ok_text_visible`, the 13-fact v2 LoRA adapter reduced the critical false positive count from 15 to 4, with the combined residual errors on these two decomposed facts falling to 3 (1 on `ins_grnd_alignment_text_visible`, 2 on `ins_ok_text_visible`). This is not simply a change of labels; it is a measurable improvement—from 13 critical false positives to 3 on the same underlying visual content—that demonstrates the effect of applying the design principle in practice.

The `ins_go` $\rightarrow$ `{ins_ok_text_visible, ins_grnd_alignment_text_visible, hsi_map_layer_vis}` decomposition is not the only example of this principle. The v1 fact `fcs_bit_result_visible` was similarly a compound target that asked the VLM to recognise a final test result rather than the visual indicators that characterise it. In v2, this is decomposed into `fcsmc_intermediate_result_visible`, `fcsmc_in_test_visible`, and `fcsmc_final_go_result_vis`, each capturing a visually distinguishable stage in the FCS-MC test sequence. The downstream procedure engine can then interpret the sequence of these facts — together with telemetry state — to determine whether the FCS BIT is complete.

The general design principle can be stated as follows: if a human annotator cannot determine the fact label by looking at a single image without knowing the procedure step, the fact is likely too high-level and should be decomposed. This principle is operationalisable and transferable to other procedures. While demonstrated on cockpit displays, this principle applies to any VLM-based perception system that must distinguish visual presence from semantic state—for example, medical imaging (tissue visible vs. condition diagnosed), industrial inspection (gauge visible vs. value in range), and document understanding (signature field visible vs. form completed).

7.4.3 Residual Failure Modes and Their Implications

Despite the ontology redesign, completion-type facts remain the most challenging category. On the Run-004 random holdout, all 8 residual critical false positives for the Qwen LoRA adapter are on `fcsmc_final_go_result_visible`. This fact requires the model to recognise a specific numeric readout (the final GO result) and to distinguish it from visually similar intermediate numeric readouts that appear in the same screen region.

There are two plausible interpretations of this residual difficulty. One is that it is a data coverage problem: the number of training examples in which the final GO result is visible is small, and the visual distinction between intermediate and final readouts may not be sufficiently represented. The other is that it is a fundamental resolution limitation: at the image resolution used for training and inference, the difference between an intermediate numeric value and the final GO value may be genuinely below the discriminative threshold of the model. Disambiguating these two interpretations would require a controlled experiment that varies image resolution and training-set composition independently, which is beyond the scope of the current work.

7.4.4 Implications for Future Ontology Design

The experience of designing two generations of visual fact ontology for the F/A-18C cold-start suggests three guidelines for ontology construction in related procedural domains:

1. **Begin with a task analysis** that identifies which procedure states require visual confirmation and which can be covered by telemetry or rule-based inference. The ontology should only contain facts for states that genuinely require visual evidence.
2. **Enforce a single-frame verifiability test** during ontology design: every fact should be evaluable from a single cockpit image without procedural context. Facts that fail this test should be decomposed or deferred to downstream reasoning.
3. **Collect composition-rebalanced data** early in the process, particularly hard negatives that expose the boundary between visually similar but procedurally distinct states. The Run-005 composition rebalance was added specifically to address under-represented fact states, and its inclusion in the training recipe was critical to the reduction in hallucination.

7.5 Limitations

The following limitations bound the scope of conclusions that can be drawn from the present work. They are stated explicitly so that the contribution can be evaluated within its proper scope and so that future work can address them systematically.

7.5.1 Limitations of the Experimental Setup

Single procedure. All experiments, benchmarks, and dataset curation efforts focus on a single procedure: the F/A-18C cold-start. There is no evidence that the 13-fact ontology, the training recipe, or the LoRA adaptation approach generalises to other procedures (e.g., taxi, takeoff, landing, or emergency procedures in DCS, or procedural tasks in other simulators). The ontology design principles articulated in Section 7.4 are plausible but untested beyond the cold-start domain.

Single composite-panel layout. The visual fact extractor is trained and evaluated on a fixed three-display composite panel (left DDI, AMPCD, right DDI). Cockpit configurations that differ in the number, arrangement, or type of displays (e.g., the F-16C with two MFDs, or the A-10C with a different instrument layout) would require retraining with new data under a new layout description. The current models have not been tested on such configurations.

Small dataset sizes. The largest training configuration uses 928 rows (Run-003 + Run-005 $\times$ 2, bilingual). This is small by comparison with typical VLM fine-tuning datasets, which often use tens or hundreds of thousands of examples (e.g., LLaVA instruction tuning and broader instruction-tuned VLM recipes) [5, 15]. While the results demonstrate that 928 rows can produce meaningful improvements, they do not establish the upper bound of what is achievable with more data. Conversely, the results do not establish the minimum data requirement; it is possible that comparable performance could be achieved with fewer examples, which would be practically relevant for domains where annotated data is scarce.

Holdout sets from the same aircraft and cockpit generation. The holdout sets (Run-002 and Run-004) are independent sessions with verified zero overlap with the training data, but they are drawn from the same aircraft type, the same cockpit configuration, and the same rendering engine (DCS World). The benchmarks therefore measure *within-domain* generalisation (new sessions for the same cockpit), not *cross-domain* generalisation (new cockpit types or new simulators).

7.5.2 Limitations of the System

VR runtime operational; pilot completed but controlled study pending. The VR runtime has been exercised in a formative pilot study with nine participants (Section 6.3), confirming that the tutoring system functions end-to-end in live VR operation. However, the pilot was small ($n = 9$), non-randomised, and descriptive. In-headset overlay rendering and latency characteristics under study conditions have been observed but not systematically measured.

No controlled human-subject evaluation. The formative pilot provides descriptive evidence of feasibility but does not constitute a controlled evaluation. The pilot provides no evidence about whether the tutoring system improves learning, retention, transfer, or workload relative to baseline conditions. The causal chain from tutoring system use to improved human learning outcomes remains untested. A larger controlled study with randomised assignment, pre-registered analysis, and adequate statistical power is required to establish pedagogical value.

The benchmark measures fact extraction, not tutoring quality. The VLM benchmarks reported in Section 6.1 are perception metrics computed on isolated single-frame fact extraction. They do not measure the quality, usefulness, or timeliness of the tutor’s help responses, the correctness of overlay targets, or the coherence of multi-turn help sequences. The pilot study addresses some of these gaps at a formative level, but an end-to-end evaluation of the full help cycle under controlled conditions has not been conducted.

7.5.3 Limitations of the Pilot Study

Small sample and non-random assignment. The pilot’s nine participants were assigned to conditions based on scheduling availability, not randomisation. With four with-tutor and five without-tutor participants, condition sizes are small and unbalanced. No inferential statistical tests are appropriate.

Variable participant experience. Participants ranged from novice to frequent DCS users. Prior experience was not balanced across conditions, and the small sample precludes statistical control for experience effects.

Incomplete workload and questionnaire data. P02’s questionnaire file was mislabelled and could not be confidently attributed; workload, TLX, and confidence data are therefore incomplete. NASA-TLX integration was not yet implemented in the export pipeline for these trials.

Without-tutor duration not comparable. Without-tutor trials ended at different procedure points (19–29 steps completed), so observed durations do not measure completion time and cannot be compared with with-tutor durations.

Passive visual verification limitation. For without-tutor trials, step completion was inferred from passive telemetry gates and action timeline review, without active VLM visual confirmation. Steps requiring visual evidence (S08, S18) may be misclassified as completed or not completed based on incomplete telemetry evidence alone.

7.5.4 Limitations of the VLM Adaptation Results

Residual critical false positives on completion-type facts. As discussed in Section 7.4, the Qwen LoRA adapter retains critical false positives on `fcsmc_final_go_result_visible` (1 on Run-002, 8 on Run-004). In a runtime deployment, each such false positive carries the risk that the tutoring system incorrectly concludes that the FCS-MC test is complete and advances the procedure prematurely. Mitigation strategies — such as the sticky-fact TTL mechanism and the requirement for multiple confirming observations — exist but have not been evaluated end-to-end.

Limited model diversity. Two backbone models were evaluated (Qwen3.5-9B and Gemma-4-31B), both with LoRA fine-tuning using the Unsloth + PEFT + TRL stack. The results do not generalise to other model architectures (e.g., LLaVA-style models [15]), other fine-tuning methods (e.g., full fine-tuning, QLoRA with different rank configurations), or other training frameworks.

No ablation of training components. The training recipe combines several design choices — bilingual SFT, Run-005×2 oversampling, composition rebalancing, LoRA rank and alpha settings — without isolating the contribution of each. It is therefore unclear which components are necessary and which are incidental. A systematic ablation study would clarify the design space but has not been conducted.

7.5.5 Scope Summary

These limitations bound the claims of this thesis. Within these bounds, the thesis demonstrates: (1) a working dual-platform tutoring runtime with automated tests and harness-level evidence validation; (2) a reproducible VLM adaptation pipeline achieving 0.99 fact accuracy on holdout data with cross-backbone evidence; (3) a pack-driven architecture supporting extension to new procedures and simulators; and (4) formative VR pilot evidence ($n = 9$) that the tutor was associated with full procedural comple-

tion and fewer coded omission errors, while assistance-coupled errors persisted. These contributions do not depend on claims that exceed the stated bounds.

7.6 Future Work

The limitations enumerated above define a concrete research agenda. This section outlines the principal directions for future work, ordered from the most immediate technical extensions to the longer-term thesis objectives.

7.6.1 Controlled Human-Subject Evaluation

The formative pilot study (Section 6.3) demonstrated feasibility and instrumentation readiness but did not provide controlled evidence about learning, workload, or retention. The next research stage is a larger controlled between-subjects evaluation comparing the grounded tutor against an appropriate baseline condition (e.g., without-tutor or static checklist). This requires completing the participant/session metadata schema, NASA-TLX integration, quiz linkage, and anonymisation pipeline; recruiting an adequate sample with randomised assignment; and pre-registering the analysis plan. Given the pilot’s observed effect on completion rates, a two-condition design (with-tutor vs. without-tutor) with $n \geq 15$ per condition would provide a more feasible starting point than the original three-condition proposal design.

7.6.2 Extending the Visual Fact Ontology to Other Procedures

The current 13-fact ontology is specific to the F/A-18C cold-start. Extending the approach to other procedures — within DCS (taxi, takeoff, landing, aerial refuelling, emergency procedures) or in other simulators — would test the generality of the ontology design principles proposed in Section 7.4. Each new procedure would require a task analysis to identify vision-priority steps, a new fact ontology designed under the single-frame verifiability constraint, capture and annotation of new training data, and independent holdout evaluation. Whether the data recipe (bilingual SFT, composition rebalancing, moderate oversampling) transfers to procedurally different domains is an open empirical question.

7.6.3 Multi-Frame Temporal Reasoning

The current visual fact extraction operates on single composite-panel images. Several cockpit states of interest involve temporal dynamics that cannot be disambiguated from a single frame: countdown values on the INS alignment page, animation sequences on the FCS-MC test display, and transient warning indicators. Extending the VLM input to short frame sequences (e.g., 3–5 consecutive frames at 1–2-second intervals) could enable the model to capture these temporal cues directly. This extension would require changes to the data capture pipeline (recording frame sequences rather than isolated screenshots), the training format (video or multi-image inputs), and the benchmark protocol (temporal fact definitions). Whether current small-scale VLMs can effectively exploit temporal information for cockpit state recognition is uncertain. Whether multi-frame VLM input is feasible at the 9B-parameter scale under LoRA training has not been tested.

7.6.4 Online Adaptation from In-Situ Corrections

The current VLM adaptation pipeline is an offline process: data is captured, pre-labeled, reviewed, and used to train a LoRA adapter in a batch cycle. In a deployed tutoring system, a learner or instructor may correct the system’s visual fact predictions in real time (e.g., “the OK text is not actually visible yet”). Collecting these in-situ corrections and using them for online or incremental model adaptation — without requiring a full re-labeling and re-training cycle — could enable the system to improve its extraction accuracy over the course of a single training session or across multiple learners. This direction requires addressing challenges in label reliability (learner corrections may themselves be erroneous), catastrophic forgetting (online updates should not degrade performance on previously correct facts), and evaluation (online adaptation must be benchmarked against a fixed holdout set that remains untouched by the adaptation process). Whether the current Unsloth + PEFT + TRL stack supports incremental fine-tuning without full retraining has not been evaluated.

7.6.5 System-Level End-to-End Evaluation

Beyond the component-level VLM benchmarks reported in this thesis, a system-level evaluation of the full help cycle is needed. This would measure, under replay or live conditions: the end-to-end latency from help trigger to overlay rendering; the correctness of overlay targets on vision-priority steps; the rate of deterministic fallback (when model

output fails validation); and the perceived usefulness of help responses as judged by domain experts reviewing replay logs. Such an evaluation would bridge the gap between the component-level benchmarks (reported here) and the pilot study (Section 6.3), providing evidence about whether the system’s components function correctly when integrated.

7.6.6 Summary of Future Directions

Table 7.1 summarises the principal future directions and their current status.

Table 7.1: Summary of future work directions.

Direction	Current status	Key challenge
Controlled human-subject evaluation	Formative pilot completed ($n = 9$); controlled study pending	Recruitment, randomisation, statistical power
Ontology extension to other procedures	13-fact v2 validated for cold-start only	Task analysis, annotation effort, domain generality
Multi-frame temporal reasoning	Single-frame pipeline operational	VLM capacity, training data volume, benchmark design
Online adaptation from corrections	Not implemented	Label reliability, catastrophic forgetting, evaluation
System-level end-to-end evaluation	Not conducted	Defining meaningful system-level metrics

The work presented in this thesis provides the technical foundation for each of these directions. The architecture, data pipeline, training infrastructure, and benchmark protocol are designed to be extensible, and the lessons learned from the two-generation ontology evolution and the cross-backbone comparison provide evidence-based guidance for the next stages of the project.

8 Conclusion

This thesis set out to design, implement, and technically validate a telemetry-grounded tutoring runtime for procedural learning in a high-fidelity flight simulator, and to gather formative evidence about its use in immersive VR training. The three contributions delivered are: an evidence-grounded tutoring architecture, a visual fact ontology and small-data VLM adaptation pipeline with cross-backbone benchmark evidence, and a VR evaluation framework exercised in a formative pilot study ($n = 9$). This chapter summarises the completed work against the three research questions, restates the contributions, and outlines directions for future work.

8.1 Summary of Completed Work

The thesis has produced three contributions, corresponding to the three research questions defined in Chapter 1.

C1 — An evidence-grounded procedural tutoring architecture. The tutoring runtime enforces that every overlay target and step recommendation carries a traceable evidence reference (telemetry variable, gating rule, RAG snippet, or visual fact), enforced at the schema level through runtime-generated JSON Schema with **if/then** rules—not merely logged for post-hoc audit. The architecture follows a ports-and-adapters pattern with a deterministic fallback path, a step harness that validates and repairs model outputs before they reach the simulator, and a procedure pack format that expresses all procedure logic as declarative YAML configuration. The architecture is supported by automated tests, a replay evaluation suite (5 cases), and 20 live-fixture regression entries covering correct advancement, safe rejection, and interaction semantics. In the VR pilot, the architecture processed 77 help cycles with zero overlay rejections.

C2 — A visual fact ontology and small-data VLM adaptation pipeline. The 13-fact v2 ontology decomposes cockpit display-page state into locally verifiable propositions. The ontology evolved from an 8-fact v1 baseline that mixed abstraction levels, producing systematic false positives on completion-type facts. The v2 decomposition—

breaking `ins_go` into `ins_grnd_alignment_text_visible` and `ins_ok_text_visible`, and decomposing the FCS-MC sequence into four sub-states—reduced critical false positives by 64% (13→4) on Run-002 and 89% (70→8) on Run-004. A Qwen3.5-9B LoRA adapter trained on 928 bilingual rows achieves 0.99 fact accuracy on both holdout sets. The identical data recipe transfers across three backbones (Qwen3.5-9B, Qwen3.6-27B, Gemma-4-31B) with consistent improvements over their respective base models. The ontology evolution yields a transferable design principle: VLMs should extract observable visual evidence, while procedural interpretation belongs in downstream reasoning.

C3 — A VR evaluation framework supported by a formative pilot. The thesis delivers study-ready instrumentation for evaluating grounded procedural tutoring in immersive simulation, and a formative VR pilot study ($n = 9$) that exercised this instrumentation. All four tutor-condition participants completed the full 33-step procedure; none of the five without-tutor participants did. With-tutor errors were predominantly assistance-coupled (CO/PA), while without-tutor errors were predominantly omissions. The pilot provides descriptive evidence of feasibility and instrumentation readiness, not inferential validation of learning effects. Larger controlled evaluation remains future work.

What this thesis demonstrates. The work establishes that a telemetry-grounded, evidence-constrained tutoring runtime with structured visual fact extraction is technically feasible and can be assessed through systematic benchmarks. It demonstrates that 928 rows of human-reviewed data suffice to adapt a general-purpose VLM to a domain-specific extraction task with near-ceiling accuracy on independent holdout sets. It shows that visual fact ontology design—specifically, the decomposition of compound procedural targets into locally verifiable atoms—has a measurable, quantitative impact on model reliability. The cross-backbone results confirm that the data recipe, not a specific architecture, drives the gains. The formative VR pilot provides initial descriptive evidence that full procedural completion was observed in all tutor-condition trials while omission errors were less frequent, and supports the readiness of the instrumentation pipeline for live VR data collection.

Scope and boundaries. The VLM benchmarks measure fact extraction accuracy, not pedagogical effectiveness. The VR pilot provides formative descriptive evidence ($n = 9$, non-randomised, unbalanced conditions) and does not establish learning gains, workload reduction, or retention improvement. Without-tutor observed durations are not compar-

able with with-tutor durations because trials were stopped at different procedure points. Questionnaire and workload data remain incomplete (P02 questionnaire mislabelled). Within these boundaries, the thesis delivers a technically evaluated tutoring runtime, a reproducible VLM adaptation pipeline with benchmark evidence, and formative VR pilot results that demonstrate feasibility and instrumentation readiness.

8.2 Future Work

The contributions of this thesis open several directions for future work, each of which builds directly on the completed contributions.

Controlled human-subject evaluation. The formative VR pilot study (Section 6.3) demonstrated feasibility and instrumentation readiness. The next step is a larger controlled between-subjects evaluation comparing the grounded tutor against an appropriate baseline, with randomised assignment, pre-registered analysis, and adequate sample size. Key infrastructure upgrades required include completion of the participant/session metadata schema, NASA-TLX integration, quiz linkage, and anonymisation pipeline. Given the pilot’s observed completion-rate pattern, a two-condition design (with-tutor vs. without-tutor) with $n \geq 15$ per condition provides a feasible starting point.

Ontology extension to other procedures. The 13-fact v2 ontology is specific to the F/A-18C cold-start task and its three-display composite panel layout. Extending the ontology to other procedures within DCS—such as navigation, weapon employment, or emergency checklists—would test the generality of the visual fact decomposition strategy and require new dataset capture and annotation cycles. The procedure pack format is designed to accommodate such extensions: a new pack directory with its own `vision_facts.yaml`, `vision_layout.yaml`, `step_registry.yaml`, and `telemetry_map.yaml` would define a new training scenario without changes to the pipeline tools.

Multi-frame temporal reasoning. The current VLM pipeline operates on single frame pairs (pre-trigger and trigger frames) and makes independent predictions per observation window. Incorporating temporal context across multiple frames could improve accuracy on facts that involve transient states or state transitions, such as the FCS-MC test sequence (`fcsmc_in_test_visible` $\rightarrow$ `fcsmc_final_go_result_visible`). Pos-

sible approaches include frame stacking as multi-image input to the VLM, recurrent aggregation of fact predictions over a sliding window, or a lightweight temporal model that consumes per-frame fact vectors and produces temporally consistent sequences.

Online adaptation. The current LoRA adapter is trained offline on a fixed dataset and deployed as a static model. An online adaptation loop, in which the system collects new labelled frames during live tutoring sessions (with tutor or expert verification) and periodically updates the adapter, could progressively improve accuracy on edge cases and adapt to changes in cockpit display configurations across DCS versions or aircraft variants. This would require lightweight fine-tuning infrastructure that can operate concurrently with the tutoring runtime and a quality-control mechanism for newly acquired labels.

Deterministic procedure engine enhancements. The current gating rule engine uses a v1 DSL with four operators (`var_gte`, `arg_in_range`, `flag_true`, `time_since`). Extending the DSL with additional operators (e.g., logical combinations, sequence counters, multi-variable conditions) and supporting user-defined gating functions would increase the expressiveness of procedure packs and reduce reliance on manual-step classification for edge cases.

The work completed in this thesis establishes that a telemetry-grounded, evidence-constrained tutoring runtime with structured visual fact extraction is technically feasible and can be assessed through automated benchmarks. The next phase of research must determine whether this capability translates into improved human learning in immersive simulation environments.

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9 Appendix

9.1 Full LoRA Hyperparameter Configuration

Table 9.1 lists the complete hyperparameter configuration for all three backbone models under the matched Run-003 + Run-005×2 recipe. Values are drawn from available training reports and benchmark artifacts.

Table 9.1: Complete LoRA fine-tuning hyperparameters across three backbones.

Parameter	Qwen3.5-9B	Qwen3.6-27B	Gemma-4-31B
max_seq_length	4096	4096	4096
num_train_epochs	4.0	4.0	4.0
learning_rate	2×10^{-4}	2×10^{-4}	2×10^{-4}
per_device_train_batch_size	1	1	1
gradient_accumulation_steps	4	4	4
lora_r	16	16	16
lora_alpha	16	16	16
lora_dropout	0.0	0.0	0.0
seed	3407	3407	3407
load_in_4bit	True	True	True
warmup_ratio	0.1	0.1	0.1
weight_decay	0.01	0.01	0.01
train_rows	928	928	928
eval_rows	0	0	0
Backbone-specific			
Vision LoRA	enabled	enabled	disabled
LoRA target modules	vision+lang	vision+lang	all-linear
Chat template	Qwen default	Qwen default	gemma-4
GPU memory util.	0.6	0.6	0.95
Training outcomes			
Train runtime (s)	6419	12619	8050
Train loss (final)	0.2156	0.1068	0.0474

Note: training loss values are not directly comparable across backbones due to differ-

ences in tokenizer behaviour, chat-template structure, target-sequence distribution, and model parameterisation.

9.2 8-Fact V1 Baseline Benchmark

The earliest complete benchmark uses an 8-fact LoRA adapter trained on Run-001 (180 tasks) and evaluated on the Run-002 holdout (50 samples). This line predates the 13-fact ontology and is included for historical reference.

Table 9.2: 8-fact v1 benchmark: Qwen3.5-9B Base vs. LoRA-v1 on Run-002 holdout (50 samples, EN).

Metric	Base	LoRA-v1	Δ
JSON valid rate	1.0	1.0	—
Schema valid rate	1.0	1.0	—
Fact accuracy	0.7600	0.9150	+0.1550
Macro F_1	0.4241	0.5847	+0.1605
Seen F_1	0.4714	0.8380	+0.3666
Sample exact match	0.06	0.36	+0.30
Critical false positives	13	15	+2

The LoRA adapter improves fact-level metrics substantially but increases critical false positives from 13 to 15, confirming the 8-fact v1 ontology’s structural weakness: compound targets such as `ins_go` were not visually verifiable from single frames and produced systematic false-positive errors on completion-type facts.

9.3 Dataset Fact Distributions

Table 9.3 reports per-fact `seen` counts for the two main training sources (Run-003, Run-005) and the two 13-fact holdout sets (Run-002 newfacts, Run-004 random). Counts are drawn from the corresponding `stats.json` files.

The Run-005 composition rebalance successfully improved coverage of underrepresented facts: `supt_page_visible` from 20 to 54 combined seen, `fcsmc_page_visible` from 55 to 111, and `ins_grnd_alignment_text_visible` from 58 to 130.

Table 9.3: Per-fact **seen** counts across training and holdout datasets (13-fact v2 ontology).

Fact ID	Run-003 (220)	Run-005 (122)	Run-002 new (50)	Run-
tac_page_visible	46	24	23	
supt_page_visible	20	34	5	
fcs_page_visible	30	40	13	
fcs_page_x_marks_visible	16	18	1	
bit_root_page_visible	18	43	9	
fcsmc_page_visible	55	56	9	
fcsmc_intermediate_result_visible	18	15	1	
fcsmc_in_test_visible	18	19	6	
fcsmc_final_go_result_visible	18	22	2	
hsi_page_visible	106	109	49	
hsi_map_layer_visible	79	37	27	
ins_grnd_alignment_text_visible	58	72	42	
ins_ok_text_visible	12	14	4	

9.4 CLI Command Reference

The **simtutor** package provides the following CLI commands:

- run.** Executes a simulation using a mock environment adapter with a pre-recorded scenario file and writes the event log.
- replay.** Validates an existing event log against a procedure pack, checking step ordering and state-machine consistency.
- score.** Scores an event log using the scoring engine, producing omission counts, state-violation counts, and a total error score.
- record-vlm.** Runs VLM fact extraction on a set of vision observations and records predictions for benchmarking.
- replay-bios.** Replays a raw DCS-BIOS capture through the telemetry pipeline and optionally invokes the help generation loop.
- validate.** Validates JSONL files against a named schema from the schema registry.

All commands accept model configuration overrides (**-model-provider**, **-model-name**, **-model-base-url**, **-model-timeout-s**) and a language switch (**-lang**).

9.5 Model Provider Configuration

Model access is configured through environment variables: `SIMTUTOR_MODEL_PROVIDER` (`openai_compat` | `ollama` | `stub`), `SIMTUTOR_MODEL_BASE_URL`, `SIMTUTOR_MODEL_NAME`, `SIMTUTOR_MODEL_TIMEOUT_S`, `SIMTUTOR_MODEL_API_KEY` (required for `openai_compat`), `SIMTUTOR_MODEL_ENABLE_MULTIMODAL` (toggles base64 image encoding in requests), and `SIMTUTOR_LANG` (`zh` | `en`). The live runtime validates all required configuration at startup and redacts API keys from all log output.

9.6 Replay Evaluation Suite

Table 9.4 lists the five replay evaluation cases in the `fa18c_startup_v04` suite.

Table 9.4: Replay evaluation cases.

Case ID	Expected step	Vision	Expected target
<code>noop_2min</code>	S01	unavailable	<code>battery_switch</code>
<code>batteryon_2min</code>	S02	unavailable	<code>fire_test_switch</code>
<code>Batteryon_enginGenoff_2min</code>	S01	unavailable	<code>generator_left_switch</code>
<code>fcs_reset_fcs_bit_2min</code>	S18	available	<code>fcs_bit_switch</code>
<code>ins_2min</code>	S12	available	<code>ins_mode_knob</code>

Cases with `vision_status = available` include per-frame sidecar files (`frames.jsonl`) in the corresponding DCS Saved Games directory, enabling replay of vision-dependent procedure steps without a live DCS instance.

9.7 Supplementary Architecture and Export Diagrams

The main chapters use simplified diagrams to support the research argument. Figures 9.1, 9.2, and 9.3 retain additional engineering detail for reproducibility and operational context.

9.8 VR Pilot Study — Supplementary Data

Table 9.5 lists participant experience profiles.

Table 9.6 lists the specific steps not completed by each without-tutor participant.

Harness and evidence validation

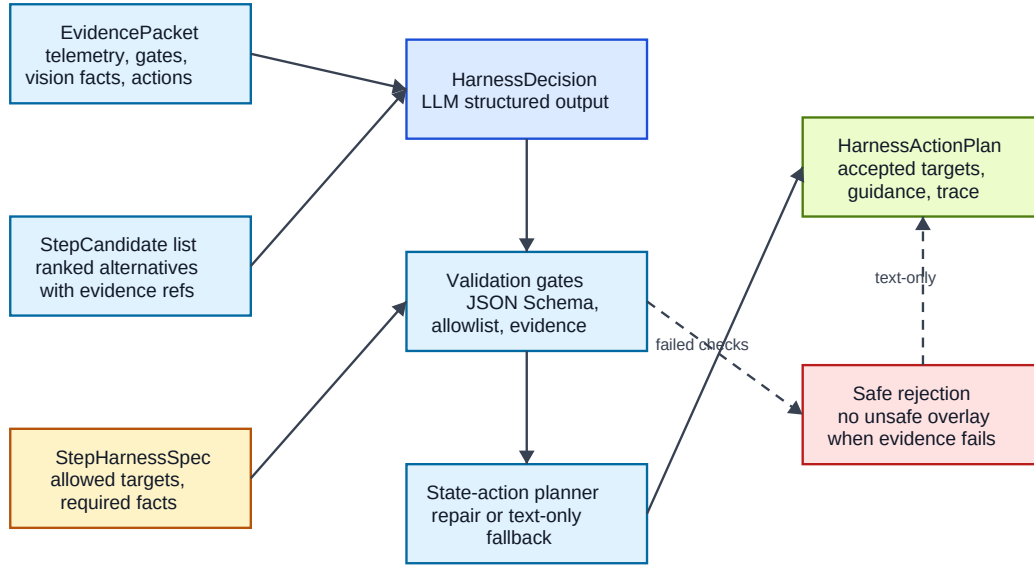


Figure 9.1: Harness and evidence validation. Candidate tutor actions are checked against schema constraints, UI allowlists, evidence consistency, and procedure state before an overlay action plan is accepted; unsafe or under-grounded actions are repaired, rejected, or downgraded to text-only guidance.

Table 9.5: Pilot participant experience profiles (self-reported).

P	Condition	DCS exp.	Flight sim.	VR exp.	Quest 3	Group
P01	without_tutor	frequent	frequent	frequent	frequent	intermediate
P02	without_tutor	intermediate	intermediate	frequent	frequent	—
P03	without_tutor	novice	novice	intermediate	frequent	beginner
P04	with_tutor	intermediate	intermediate	frequent	frequent	intermediate
P05	with_tutor	intermediate	intermediate	frequent	frequent	—
P06	without_tutor	novice	novice	intermediate	frequent	beginner
P07	without_tutor	novice	novice	intermediate	frequent	beginner
P08	with_tutor	novice	novice	frequent	frequent	—
P09	with_tutor	novice	novice	intermediate	intermediate	beginner

— = not recorded.

Deployment topology

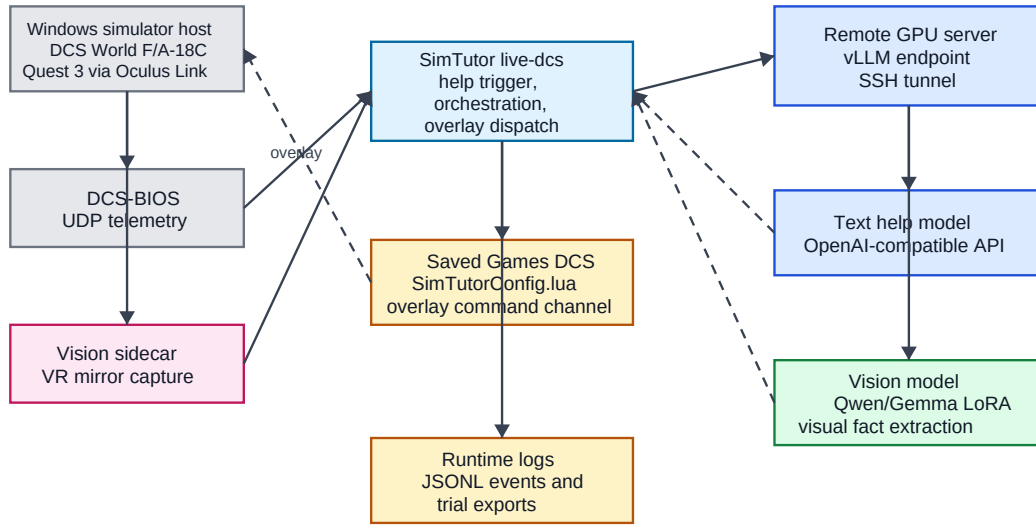


Figure 9.2: Deployment topology used by the prototype. The simulator, Quest 3 overlay path, local runtime, replay utilities, and remote model providers are separated so that live trials and offline analysis can use the same evidence and logging contracts.

Table 9.6: Steps not completed in the without-tutor condition (manual coding).

P	Steps not completed
P01	S09, S12, S18, S19, S20, S22, S24, S27, S29, S31
P02	S02, S14, S18, S19, S27, S30
P03	S07, S09, S16, S18, S19, S26, S27, S29, S32
P06	S07, S18, S19, S27
P07	S02, S09, S12, S13, S16, S17, S18, S19, S22, S26, S27, S28, S29, S32

S18 (FCS-MC BIT page on right DDI) and S19 (FCS BIT final GO result) are missed by all five without-tutor participants. S27 (flap switch to AUTO) is missed by all five. These steps are vision-priority and require either VLM confirmation or instructor knowledge of the correct display-page sequence.

Detailed experiment export pipeline

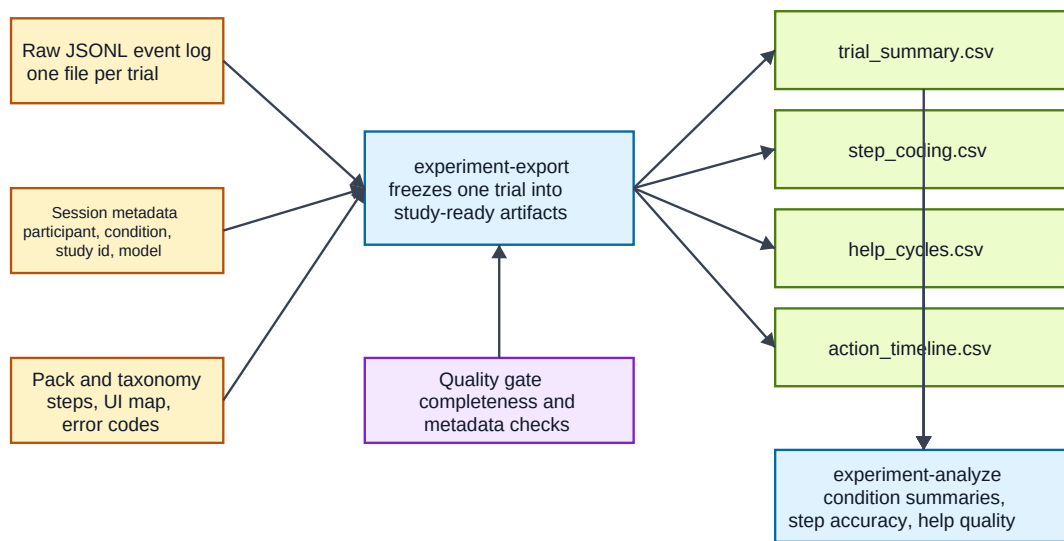


Figure 9.3: Detailed experiment export pipeline. Runtime event logs and trial metadata are transformed into participant-level summaries, step coding, help-cycle records, quality gates, and study-level analysis tables.

Table 9.7 reports the auto-coded error counts (before manual review) alongside the manual error counts, illustrating the magnitude of auto-to-manual correction in this pilot.

Table 9.7: Auto-coded vs. manual error counts per participant.

P	Condition	Auto OM	Auto CO	Manual OM	Manual CO	Total manual
P01	without_tutor	4	3	8	0	10
P02	without_tutor	0	3	0	0	1
P03	without_tutor	0	4	7	0	11
P06	without_tutor	0	3	2	0	4
P07	without_tutor	1	9	11	0	11
P04	with_tutor	0	4	0	3	7
P05	with_tutor	0	3	0	3	6
P08	with_tutor	0	3	0	0	0
P09	with_tutor	0	3	0	3	5

Auto-coded errors are from the automated pipeline (Auto_Error_* columns in `step_coding.csv`). Manual errors are from the human-reviewed columns (Error_*). P08's auto errors were all overridden to zero by manual review (see data-quality note in Section 6.3).